



BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-1 : INTRODUCTION TO SAD



1. FUNDAMENTALS OF SYSTEM

English Definition

A system is a group of interrelated components working together to achieve a common goal. It accepts input from the environment, processes it and produces output with feedback control.

Hinglish (Simple)

System matlab kai parts mil kar ek goal achieve karte hain. Ye input lete hain, process karte hain, output dete hain aur feedback se improve hote hain.

Real Life Example

Example – ATM System

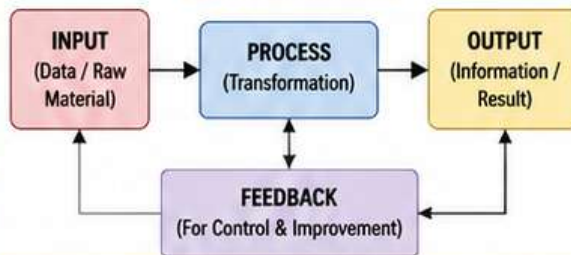
Input: Card, PIN

Process: Verify & Check Balance

Output: Cash / Balance

Feedback: SMS / Receipt

BASIC SYSTEM MODEL



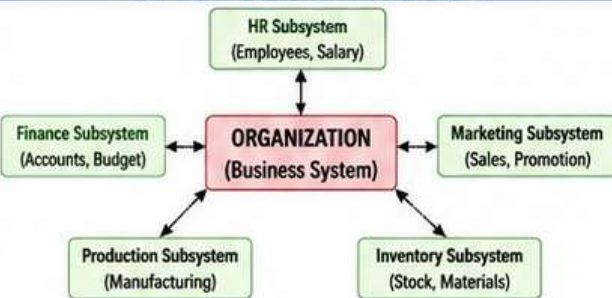
★ Important Terms

- **Input** : Data or resources given to the system.
- **Process** : Activities that convert input to output.
- **Output** : Result or information produced.
- **Feedback** : Output used to improve system.
- **Environment** : External elements affecting system.
- **Boundary** : Limits that separate system from environment.
- **Interface** : Point where system interacts.
- **Subsystem** : Smaller system within a big system.
- **Goal** : Purpose of the system.

2. CLASSIFICATION OF SYSTEMS

	Physical System	Tangible, can be seen and touched. (Example: Computer)
	Abstract System	Conceptual, cannot be seen. (Example: Theory)
	Open System	Interact with environment. (Example: Business)
	Closed System	No interaction with environment. (Example: Sealed bottle)
	Deterministic System	Output is predictable. (Example: Calculator)
	Probabilistic System	Output is uncertain. (Example: Weather)
	Manual System	Operated by humans. (Example: Typewriter)
	Computerized System	Operated by computers. (Example: Banking System)
	Natural System	Occurs in nature. (Example: Rain)
	Man-made System	Created by humans. (Example: Railway System)

3. REAL LIFE BUSINESS SUBSYSTEMS



Example – A Company has different subsystems, but all work together to achieve main goal (Profit & Growth).

4. REAL TIME SYSTEMS



Real time system gives output within a specific time limit.

Examples:

- Traffic Signal System
- Airline Reservation System
- Industrial Control System
- ICU Patient Monitoring
- UPI Payment System

5. DISTRIBUTED SYSTEMS



Distributed system has multiple independent computers working together over a network.

Examples:

- Internet
- Google Drive
- Banking Network
- Cloud Computing
- E-commerce Platforms

6. DEVELOPMENT OF A SUCCESSFUL SYSTEM



7. VARIOUS APPROACHES FOR DEVELOPMENT OF INFORMATION SYSTEMS

- 1 Waterfall Model – Step by step approach.
- 2 Prototype Model – Build prototype, test & refine.
- 3 Spiral Model – Risk driven & iterative.
- 4 Incremental Model – System built in parts.
- 5 Agile Model – Flexible & fast development.



8. STRUCTURED ANALYSIS & DESIGN APPROACH



Structured approach follows proper steps and produces well-documented & reliable system.

9. PROTOTYPE

Prototype is a sample model of the system used to understand requirements and get feedback from users.

Types:

- Throwing Prototype – Used only for requirement gathering.
- Evolutionary Prototype – Improved and becomes final system.

Example: Before making a mobile app, developer shows demo version to users.



10. JOINT APPLICATION DEVELOPMENT (JAD)

JAD is a technique where users, managers and analysts work together in structured sessions to define requirements and design system.

Benefits:

- Better Communication
- Fast Requirement Gathering
- High User Satisfaction
- Reduced Development Time



★ UNIT-1 QUICK REVISION

- ✓ System = Input → Process → Output → Feedback
- ✓ Types = Physical, Abstract, Open, Closed, Deterministic, Probabilistic, Manual, Computerized, Natural, Man-made
- ✓ Real Time = Time bound output
- ✓ Distributed = Multiple systems connected in network
- ✓ Approaches = Waterfall, Prototype, Spiral, Incremental, Agile
- ✓ Prototype = Sample model for user feedback
- ✓ JAD = Users + Analyst + Manager = Better System

★ PYQ INDICATOR (AKU 2014–2025)

- | | |
|--|------------------------|
| • Define System. Write types of system. | 2014, 2016, 2018, 2022 |
| • What is Real Time System? Give examples. | • 2015, 2019, 2021 |
| • What is Distributed System? Give examples. | • 2016, 2020, 2023 |
| • What is Prototype? Explain its types. | • 2014, 2017, 2021 |
| • What is JAD? Explain benefits. | • 2015, 2018, 2022 |
| • Explain Structured Analysis & Design Approach. | • 2013, 2017, 2023 |

★ SHORT TRICK BOX

- ★ Remember I-P-O-F
Input – Process – Output – Feedback
- ★ Remember P-A-D-T-I-M
Planning – Analysis – Design – Testing – Implementation – Maintenance
- ★ System Good = User Happy + Time Save + Cost Save + Quality Improve

★ 2-MARK READY ANSWER (ANY TOPIC)

- System is a group of components working together to achieve a common goal.
- Real time system gives output within specific time limit.
- Distributed system works on multiple computers connected through network.
- Prototype is a sample model used for user feedback.
- JAD is group approach where users and analysts work together.

ONE LINE SUMMARY: "System samajhna aur usko sahi tarike se design karna hi SAD hai."



BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-1 : INTRODUCTION TO SAD

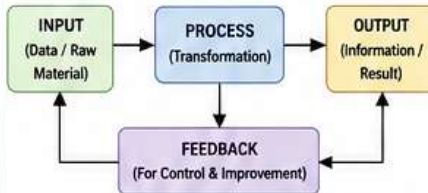


1. FUNDAMENTALS OF SYSTEM

Definition

A system is a set of interrelated components working together to achieve a common goal. It accepts input from the environment, processes it and produces output with feedback control.

Basic System Model



Characteristics of System

- ✓ Organization – System is a group of components.
- ✓ Interaction – Components interact with each other.
- ✓ Interdependence – Components depend on each other.
- ✓ Integration – All parts work together as a whole.
- ✓ Objective – Every system has a specific goal.
- ✓ Environment – System exists in an environment.
- ✓ Boundary – System has a boundary that separates it from environment.

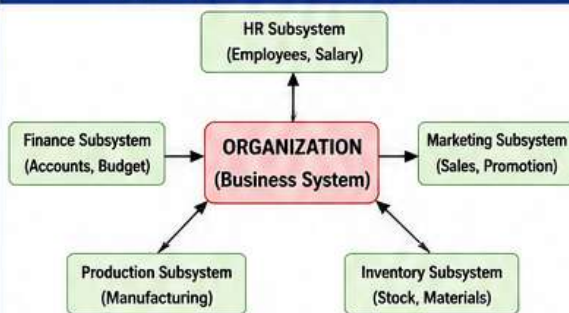
2. IMPORTANT TERMS RELATED TO SYSTEMS

	Input	Data or resources given to the system.
	Process	Activities that convert input to output.
	Output	Result or information produced by the system.
	Feedback	Output used to improve or control the system.
	Environment	External elements affecting the system.
	Boundary	Limits that separate system from environment.
	Interface	Point where system interacts with other system or environment.
	Subsystem	Smaller system within a larger system.
	Goal	Purpose or objective of the system.

3. CLASSIFICATION OF SYSTEMS

Basis	Types	Description / Example
	• Physical System	• Tangible (Computer)
	• Abstract System	• Intangible (Theory)
	• Open System	• Interact with environment (Business)
	• Closed System	• No interaction (Sealed bottle)
	• Deterministic System	• Output is predictable (Calculator)
	• Probabilistic System	• Output is uncertain (Weather)
	• Manual System	• Operated by humans (Typewriter)
	• Computerized System	• Operated by computers (Banking System)
	• Natural System	• Occurs in nature (Rain)
	• Man-made System	• Created by humans (Railway System)
	• Static System	• Do not change with time (Mathematical formula)
	• Dynamic System	• Change with time (Stock market)

4. REAL LIFE BUSINESS SUBSYSTEMS



A business system is composed of many subsystems. All subsystems work together to achieve the main goal (Profit & Growth).

5. REAL TIME SYSTEMS



Real time systems process data and give output within a specific time limit.

Examples

- Traffic Signal System
- Airline Reservation System
- Industrial Control System
- ICU Patient Monitoring System
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6. DISTRIBUTED SYSTEMS



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7. DEVELOPMENT OF A SUCCESSFUL SYSTEM

- ✓ Clear understanding of user requirements.
- ✓ Proper planning and scheduling.
- ✓ Feasibility study and cost-benefit analysis.
- ✓ Qualified and committed team.
- ✓ Effective communication.
- ✓ Quality design and development.
- ✓ Thorough testing and quality assurance.
- ✓ Proper implementation and training.
- ✓ Regular maintenance and feedback.



8. VARIOUS APPROACHES FOR DEVELOPMENT OF INFORMATION SYSTEMS

- Waterfall Model** Sequential phases. Each phase must be completed before next.
- Prototype Model** Build prototype, test and refine through user feedback.
- Spiral Model** Iterative and risk-driven model.
- Incremental Model** System is built in small increments.
- Agile Model** Flexible and fast development with continuous feedback.



9. STRUCTURED ANALYSIS AND DESIGN (SAD) APPROACH



Structured approach follows proper steps and produces well-documented and reliable system.

Features of SAD Approach

- ✓ Step-by-step methodology.
- ✓ Focus on documentation.
- ✓ Easy to understand and control.
- ✓ Ensures quality and reliability.

Advantages

- Clear phases and deliverables.
- Better planning and control.
- Easy maintenance.
- Reduced risk.

10. PROTOTYPE

Prototype is a sample model of the system used to understand requirements and get feedback from users before building the final system.

Types of Prototype

- Throwaway Prototype – Used only for requirement gathering, then discarded.
- Evolutionary Prototype – Improved and becomes final system.

Example : Before building a mobile app, developer shows demo version to users.



11. JOINT APPLICATION DEVELOPMENT (JAD)

JAD is a technique in which users, managers and analysts work together in structured sessions to define requirements and design the system.

Benefits

- Better Communication
- Fast Requirement Gathering
- High User Satisfaction
- Reduced Development Time



SUMMARY OF UNIT - 1

- ✓ System = Input → Process → Output → Feedback
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- Explain Structured Analysis & Design Approach. 2013, 2017, 2023

SHORT TRICK BOX

- ★ Remember I-P-O-F
Input – Process – Output – Feedback
- ★ Remember P-A-D-T-I-M
Planning – Analysis – Design – Testing – Implementation – Maintenance
- ★ System Goal = User Happy + Time Save + Cost Save + Quality Improve

2-MARK READY ANSWER (ANY TOPIC)

- System is a group of components working together to achieve a common goal.
- Real time system gives output within specific time limit.
- Distributed system works on multiple computers connected through network.
- Prototype is a sample model used for user feedback.
- JAD is group approach where users and analysts work together.



ONE LINE SUMMARY : A system is a set of interrelated components working together to achieve a common goal.
SAD is the structured approach to design and develop a quality information system.



BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-2 : SYSTEMS ANALYST – A PROFESSION



1. WHY DO BUSINESSES NEED SYSTEMS ANALYSTS?

- To identify business problems and opportunities.
- To convert user needs into system requirements.
- To design systems that improve efficiency and effectiveness.
- To reduce cost and increase productivity.
- To ensure proper use of resources and technology.
- To act as a bridge between users and developers.
- To support decision making with accurate information.
- To ensure successful implementation of systems.
- To adapt systems to changing business needs.



2. USERS

Users are the people who use the information system or get benefit from it.



Types of Users

- **End Users** – Use the system for their day-to-day work.
- **Operational Users** – Work with transactional data.
- **Managerial Users** – Use information for planning and control.
- **Executive Users** – Use summary information for strategic decision.

3. ANALYSTS IN VARIOUS FUNCTIONAL AREAS

Functional Area	Role of Systems Analyst
Finance & Accounting	Design systems for budgeting, accounting, financial reporting.
Marketing & Sales	Analyze customer data, sales forecasting, CRM systems.
Human Resource	Design payroll, recruitment, training and employee management systems.
Production & Operations	Improve production planning, inventory and quality control systems.
Information Technology	Involve in system architecture, development and maintenance.
... Other Areas	Support systems for legal, research, administration, etc.

4. ROLE OF A SYSTEMS ANALYST

A Systems Analyst plays a key role in the development of information systems.

Major Roles

- Study the current system and identify problems.
- Elicit and analyze user requirements.
- Design logical and physical solutions.
- Prepare documentation and reports.
- Evaluate alternative solutions.
- Coordinate with users, managers and developers.
- Ensure successful implementation and maintenance.



5. DUTIES OF A SYSTEMS ANALYST

- ✓ Investigate and understand the existing system.
- ✓ Collect and analyze requirements.
- ✓ Prepare system requirement specification (SRS).
- ✓ Develop system models and designs.
- ✓ Evaluate hardware and software alternatives.
- ✓ Prepare cost-benefit analysis.
- ✓ Prepare test plans and evaluate system.
- ✓ Monitor system implementation.
- ✓ Train users and provide support.
- ✓ Maintain and improve systems.



6. QUALIFICATIONS OF A SYSTEMS ANALYST

- Bachelor's degree in Computer Science / Information Technology / related field.
- Knowledge of systems development life cycle.
- Understanding of business processes.
- Good academic record.
- Relevant certifications (advantage).
- Continuous learning and updated technology knowledge.



7. SKILLS REQUIRED BY A SYSTEMS ANALYST

A. ANALYTICAL SKILLS

- Ability to study and break down complex problems.
- Identify cause-effect relationships.
- Evaluate alternatives.
- Logical thinking and decision making.



B. TECHNICAL SKILLS

- Knowledge of computer systems, databases, networks.
- Understanding of programming concepts.
- Familiarity with software development tools.
- Knowledge of system analysis methods and models.



C. MANAGEMENT SKILLS

- Planning and organizing work.
- Resource management.
- Time management.
- Budget estimation and control.
- Risk management.
- Leadership quality.



D. INTERPERSONAL SKILLS

- Good communication (oral & written).
- Active listening.
- Able to work in team.
- Build relationships.
- Conflict resolution.
- Negotiation skills.



8. COMPARISON: USERS vs SYSTEMS ANALYST

Basis	Users	Systems Analyst
Focus	Use the system for daily work	Design and improve the system
Knowledge	Know about business process	Know about system development
View	Short-term and operational	Long-term and strategic
Responsibility	Use system effectively	Ensure system is effective and efficient
Involvement	Provide requirements and feedback	Analyze, design and deliver solutions
Goal	Complete their work	Improve business through better systems

9. KEY SUCCESS FACTORS FOR A SYSTEMS ANALYST

- ★ Strong analytical ability
- ★ Good communication
- ★ Business understanding
- ★ Technical knowledge
- ★ Teamwork and leadership
- ★ Adaptability and flexibility
- ★ Professional attitude
- ★ Ethical behavior

10. CAREER OPPORTUNITIES

- Business Systems Analyst
- System Analyst
- Requirements Analyst
- Functional Analyst
- IT Consultant
- Project Manager
- Product Owner
- Data Analyst (related role)



11. ONE LINE SUMMARY

A Systems Analyst understands business needs, designs effective systems and works as a bridge between users and technology to achieve organizational goals.



PYQ INDICATOR (2014 – 2025)

- Explain the role of the system analyst.
- Duties of a systems analyst.
- Qualities/skills required by a systems analyst.
- Why do businesses need systems analysts?
- Analysts in various functional areas.
- Interpersonal skills of systems analyst.



SHORT TRICK BOX

- ★ Analytical, Technical, Management, Interpersonal = "A-T-M-I"
- ★ Duties Easy Remember:
I-C-A-D-E-T-M-T-S-S
(Investigate, Collect, Analyze, Design, Evaluate, Test, Monitor, Train, Support)

2-MARK READY ANSWER (ANY TOPIC)

- ✓ Role = Understand, Design, Coordinate, Implement
- ✓ Duties = Analyze, Design, Document, Implement, Support
- ✓ Skills = Analytical, Technical, Management, Interpersonal
- ✓ Need = Efficiency, Accuracy, Cost Control, Decision Making
- ✓ Users = End, Operational, Managerial, Executive



ONE LINE FINAL SUMMARY : A Systems Analyst is a problem solver who improves business performance by designing and implementing effective information systems.





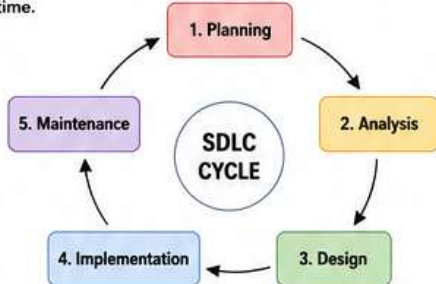
BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-3 : PROCESS OF SYSTEM DEVELOPMENT



1. SYSTEMS DEVELOPMENT LIFE CYCLE (SDLC)

SDLC is a structured process used to design, develop, implement and maintain information systems. It provides a framework to build quality systems within budget and time.



2. PHASES OF SDLC

- 1 Project Identification & Selection
- 2 Project Initiation & Planning
- 3 Analysis
- 4 Logical Design
- 5 Physical Design
- 6 Implementation
- 7 Maintenance



💡 SDLC is an iterative process. Feedback from maintenance may lead back to earlier phases.

3. PROJECT IDENTIFICATION AND SELECTION

Identification

Recognizing problems or opportunities where information systems can help the organization.



Selection

Evaluating identified projects and choosing the one that provides maximum benefit with available resources.



Evaluation Criteria

- ✓ Strategic fit
- ✓ Feasibility
- ✓ Cost
- ✓ Risk
- Expected benefits
- Resources availability
- Legal & ethical issues
- Priority

4. PROJECT INITIATION AND PLANNING

Project Initiation

- Formulate the project idea into a formal proposal.
- Define project scope, objectives and major deliverables.
- Identify stakeholders and their responsibilities.
- Obtain management approval.

Project Planning

- Define detailed activities and tasks.
- Estimate time, cost and resources.
- Prepare project schedule (Gantt chart).
- Identify risks and prepare mitigation plans.
- Establish monitoring and control mechanisms.



Deliverables

- Project Charter
- Scope Statement
- Feasibility Report
- Project Plan
- Resource Plan
- Cost Estimate
- Schedule Plan
- Risk Management Plan



5. ANALYSIS

The analysis phase focuses on understanding the current system and identifying user requirements.

Activities

- Study current system.
- Identify problems and opportunities.
- Gather requirements from users.
- Analyze data, processes and controls.
- Evaluate feasibility.
- Prepare System Requirements Specification (SRS).



Tools & Techniques

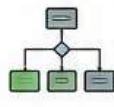
- Interviews
- Questionnaires
- Observation
- Document Review
- Sampling
- JAD / Workshops

6. LOGICAL DESIGN

Logical design converts user requirements into a logical system specification independent of hardware and software.

Activities

- Design system architecture.
- Design processes and data.
- Create Data Flow Diagrams (DFD).
- Design database (ER Diagram).
- Design inputs, outputs and user interface logically.
- Prepare logical data dictionary.



7. PHYSICAL DESIGN

Physical design converts the logical design into a physical system specification.

Activities

- Select hardware and software.
- Design database physically.
- Design file structures.
- Design input/output formats.
- Design user interface.
- Prepare physical data dictionary.



8. IMPLEMENTATION

Implementation is the phase where the designed system is built and put into use.

Activities

- Develop / Acquire software.
- Install hardware and software.
- Test the system.
- Convert data.
- Train users.
- Changeover (Direct / Parallel / Phased / Pilot).
- Go live and support users.



9. MAINTENANCE

After implementation, the system must be maintained to ensure continuing effectiveness.

Types of Maintenance

	Corrective Maintenance	Fixing errors or problems that occur in the system.
	Adaptive Maintenance	Modifying system due to changes in environment.
	Perfective Maintenance	Improving performance or adding new features.
	Preventive Maintenance	Making changes to prevent future problems.

10. PRODUCT (DELIVERABLES) OF SDLC PHASES

Phase	Major Deliverables
1. Identification & Selection	Problem Statement, Project Proposal, Feasibility Study
2. Initiation & Planning	Project Plan, Schedule, Budget, Resource Plan
3. Analysis	SRS Document, Feasibility Report, Process Models
4. Logical Design	DFD, ER Diagram, Logical Data Dictionary, Process Specs
5. Physical Design	Database Design, File Structures, I/O Design, Physical Data Dictionary
6. Implementation	Working System, Test Reports, User Training, Documentation
7. Maintenance	Maintenance Reports, Change Requests, Updated System

11. APPROACHES TO DEVELOPMENT

- **Waterfall Model** – Linear and sequential approach.
- **Prototyping** – Build prototype, get feedback, refine and develop final system.
- **Incremental Model** – Build system in small increments.
- **Spiral Model** – Risk-driven, iterative cycles.
- **Agile Model** – Iterative, flexible and customer collaboration based.



12. PROTOTYPING

Prototyping is a technique where a working model (prototype) is created to understand requirements better.

Benefits

- Better understanding of user requirements.
- Early feedback.
- Reduces risk.
- Improves accuracy of final system.

Prototype Process



13. JOINT APPLICATION DESIGN (JAD)

JAD is a structured session where users, managers, and analysts work together to define requirements and design systems.

Benefits

- Better communication.
- Fast requirements gathering.
- High user satisfaction.
- Reduced development time.



14. PARTICIPATORY DESIGN

Users actively participate throughout the development process.

Benefits

- Users feel ownership.
- System meets real needs.
- Better acceptance.
- Improved quality.



PYQ INDICATOR (2014 - 2025)

- ★ Explain phases of SDLC.
- ★ What are the activities of SDLC?
- ★ What is logical design?
- ★ What is feasibility study?
- ★ Explain prototyping.
- ★ What is JAD? Explain its benefits.
- ★ Why maintenance is important?



SHORT TRICK BOX

- ★ Remember SDLC Phases: P-A-D-I-M
Planning – Analysis – Design – Implementation – Maintenance
- ★ Logical vs Physical Design
Logical = What system will do?
Physical = How system will do?
- ★ Prototyping = Sample model for user feedback.

12-MARK READY ANSWER (ANY TOPIC)

- ✓ SDLC has 7 phases.
- ✓ Logical design is technology independent.
- ✓ Physical design is technology dependent.
- ✓ Prototyping uses sample model.
- ✓ JAD involves users + analysts + managers.
- ✓ Maintenance ensures system works properly.

ONE LINE SUMMARY

SDLC is a complete cycle of planning, analyzing, designing, implementing and maintaining systems to meet user needs effectively and efficiently.



UNIT-3 FINAL ONE LINE: SDLC provides a systematic process to develop quality information systems through proper planning, analysis, design, implementation and maintenance.





BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-4 : INTRODUCTION TO DOCUMENTATION OF SYSTEMS



1. CONCEPTS AND PROCESS OF DOCUMENTATION

Concept:

Documentation is the process of recording information about the system, its requirements, design, development, testing, implementation and maintenance in a clear and systematic manner.

Process:

- 1. Planning Documentation**
Decide what to document, for whom and how.
- 2. Collect Information**
Gather data from users, analysts, designers, etc.
- 3. Prepare Documents**
Write documents using proper formats and standards.
- 4. Review & Verify**
Check accuracy, completeness and consistency.
- 5. Approve & Maintain**
Get approval and keep documents updated.

2. TYPES OF DOCUMENTATION

Type	Purpose
1. Planning Documents	Define scope, objectives, schedule, resources.
2. Requirement Documents	Describe user needs and system requirements.
3. Design Documents	Explain system design (logical & physical).
4. Development Documents	Include code, program logic, technical details.
5. Testing Documents	Plan and record testing activities and results.
6. Operation Documents	Help in running the system (user manual).
7. Maintenance Documents	Help in future changes and maintenance.

Good documentation supports the system throughout its life cycle.

3. SYSTEM REQUIREMENTS SPECIFICATION (SRS)

Definition:

SRS is a document that describes what the system should do, not how it will do it.

Contents:

- Introduction
- Overall description of system
- Functional requirements
- Non-functional requirements
- Constraints
- Assumptions
- External interfaces
- Acceptance criteria

SRS

Importance:

- Acts as a contract between users and developers.
- Basis for design, development and testing.
- Reduces misunderstanding.
- Helps in change management.

4. SYSTEM DESIGN SPECIFICATION (SDS)

Definition:

SDS describes how the system will meet the requirements defined in the SRS.

Contents:

- System architecture
- Data design
- Process design
- Input/Output design
- Database design
- File structures
- Security and control design
- Interface design

SDS



Importance:

- Provides a blueprint for developers.
- Guides coding and database creation.
- Helps in maintenance and enhancement.

5. TEST DESIGN DOCUMENT

Definition:

It describes the testing approach, test cases, procedures and expected results.

Contents:

- Test objectives
- Test strategy
- Test environment
- Test data
- Test cases and procedures
- Expected results
- Entry and exit criteria

TEST PLAN



Importance:

- Ensures complete and systematic testing.
- Helps in identifying defects early.
- Provides traceability to requirements.

6. USER MANUAL

Definition:

A user manual is a document that helps end users to understand and use the system effectively.

Contents:

- Introduction
- System overview
- Hardware/Software requirements
- Installation instructions
- Step-by-step user instructions
- Screenshots/Examples
- Troubleshooting
- FAQs and Tips

USER MANUAL



Importance:

- Helps users to operate the system correctly.
- Reduces training time.
- Improves user satisfaction.

7. DIFFERENT STANDARDS FOR DOCUMENTATION

Standard	Full Form	Purpose
IEEE 830	IEEE Recommended Practice for Software Requirements Specifications	Guidelines for writing SRS.
IEEE 1016	IEEE Standard for Software Design Descriptions	Guidelines for writing design documents.
ISO/IEC 12207	Information Technology – Software Life Cycle Processes	Provides process framework and documentation needs.
ISO/IEC 9126	Software Product Evaluation – Quality Characteristics	Defines quality attributes for documentation.
IBM Style Guide	IBM Documentation Style Guide	Standard format for technical documents.

8. DOCUMENTATION AND QUALITY OF SOFTWARE

Good documentation contributes to software quality by:

- Ensuring correctness and completeness.
- Improving maintainability.
- Supporting testing and quality assurance.
- Facilitating communication among team members.
- Helping in training new users and developers.
- Reducing errors and rework.



Quality Characteristics Improved:

- Reliability
- Maintainability
- Usability
- Efficiency
- Portability
- Understandability

9. GOOD PRACTICES FOR DOCUMENTATION

- Be Clear and Simple** – Use easy language and short sentences.
- Be Consistent** – Maintain uniform format, terms and style.
- Be Complete** – Include all necessary information.
- Be Accurate** – Ensure correctness and up-to-date content.
- Use Standard Templates** – Follow standard formats and guidelines.
- Use Diagrams and Charts** – Use visuals for better understanding.
- Update Regularly** – Keep documents updated with changes.
- Store Properly** – Maintain version control and easy access.
- Ensure Security** – Protect documents from unauthorized access.
- Review Periodically** – Check and improve documents regularly.

Remember:

"If it is not documented, it does not exist."



10. DOCUMENTATION DELIVERABLES AT A GLANCE

Document	Main Deliverable
SRS	What the system should do.
SDS	How the system will be built.
Test Design Document	How the system will be tested.
User Manual	How users will operate the system.
Technical/Programmer Manual	How developers will maintain the system.
Operation Manual	How system is run and controlled.
Installation Guide	How system is installed.
Maintenance Manual	How system is maintained and updated.

PYQ INDICATOR (AKU 2014 – 2025)

- Explain different types of documentation.
- What is SRS? Write its contents.
- Describe the importance of documentation.
- What is SDS? How is it useful?
- Write short notes on User Manual.
- List the standards of documentation.

SHORT TRICK BOX

SRS – WHAT
SDS – HOW
TEST DOC – HOW TO TEST
USER MANUAL – HOW TO USE



2-MARK READY ANSWER (ANY TOPIC)

- Documentation is the process of recording information about system.
- SRS describes what system should do.
- SDS describes how system will be built.
- Test document describes how system is tested.
- User manual helps end users to operate system.

ONE LINE SUMMARY

Documentation is the backbone of software development that ensures clarity, quality, control and successful delivery of information systems.



UNIT-4 FINAL ONE LINE: Good documentation makes a good system better and a better system excellent.





BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-5 : PLANNING AND DESIGNING SYSTEMS



1. PROCESS OF SYSTEM PLANNING

System planning is the process of deciding what information system is needed, why it is needed and how it will be developed.

Steps in System Planning



2. FACT FINDING TECHNIQUES

Fact finding means collecting relevant information about the current system.

Technique	Purpose
Interviews	One-to-one discussion to get information.
Group Discussion	Discuss with a group of users for ideas.
Site Visits (Observation)	Visit the work place and observe activities.
Presentations	Present ideas and get feedback from users.
Questionnaires	Written set of questions sent to users.
Sampling of Documents	Study existing documents, forms, records, reports.

3. FEASIBILITY STUDY

Feasibility study is done to check whether the proposed system is practical, economical and beneficial.

Types of Feasibility

Technical Feasibility	Checks if required technical resources, hardware, software and technology are available.
Operational Feasibility	Checks if system will be accepted by users and can be used easily.
Economic Feasibility	Checks if benefits of system are more than its cost.
Legal Feasibility	Checks if system follows legal rules, policies and regulations.

A feasible system should be:

✓ Practical ✓ Acceptable ✓ Affordable ✓ Legal

4. COST-BENEFIT ANALYSIS

It is a technique to compare total cost of the system with total benefits.

Costs	Benefits
• Hardware Cost • Software Cost • Development Cost • Training Cost • Maintenance Cost • Operating Cost	• Increase in Profit • Better Service • Time Saving • Accuracy • Better Control • Competitive Advantage



Decision Rule: If Total Benefits > Total Costs, then the project is economically feasible.

5. PREPARING SCHEDULE

A schedule shows when each activity will start and finish.

Tools Used

- Gantt Chart
- PERT Chart
- Network Diagram



Gantt Chart (Example)

Activity	Week 1	Week 2	Week 3	Week 4
Analysis	■			
Design		■	■	
Development			■	■
Testing				■
Implementation				■

6. GATHERING REQUIREMENTS OF SYSTEM

Collect all details about what the system should do.

Types of Requirements

- **Functional Requirements** – What the system should do.
- **Non-functional Requirements** – How well the system should do.

Sources of Requirements

- Users
- Managers
- Existing Documents
- Observation
- Surveys / Questionnaires

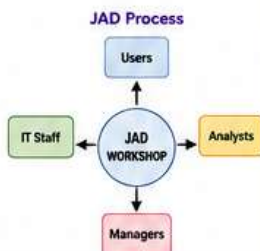


7. JOINT APPLICATION DEVELOPMENT (JAD)

JAD is a structured approach where users, managers and analysts work together in workshops to define requirements and design system.

Benefits

- Better communication
- Fast decision making
- High user satisfaction
- Reduced development time
- Better quality system



8. PROTOTYPING

Prototyping is building a sample model (prototype) of the system to understand requirements better.

Reasons

- Clarifies requirements
- Early feedback from users
- Reduces risk
- Improves accuracy

Prototype Process



Types of Prototype

- **Throwaway Prototype**
Used only for requirement gathering.
- **Evolutionary Prototype**
Improved and becomes final system.



9. ISSUES INVOLVED IN FEASIBILITY STUDY

- Lack of accurate information
- Resistance to change
- High initial cost
- Lack of technical skills
- Time constraints
- Uncertainty of future benefits
- Legal and security issues

10. COMPARISON: TECHNIQUES FOR FACT FINDING

Technique	Best Used When	Advantages	Limitations
Interview	Detailed information needed	Flexible, personal, in-depth	Time consuming, costly
Group Discussion	Ideas from many people needed	More ideas, quick results	Dominance by few people
Site Visit	Need to see actual work	Real picture, accurate	Time and travel cost
Questionnaire	Large number of users	Inexpensive, standard questions	Low response, less detail
Sampling of Documents	Existing data available	saves time, useful records	May be outdated/incomplete

11. QUICK REVISION POINTS

- ✓ System planning decides what, why and how of information system.
- ✓ Fact finding collects information using different techniques.
- ✓ Feasibility study ensures system is practical and beneficial.
- ✓ Requirements gathering identifies what users need.
- ✓ JAD and Prototyping improve user involvement and satisfaction.
- ✓ Cost-Benefit analysis helps in economic decision.
- ✓ A good schedule ensures project is completed on time.

12. ONE LINE SUMMARY

Planning and designing systems involve collecting information, analyzing feasibility, gathering requirements and preparing a proper plan to build a useful, acceptable and beneficial system.



PYQ INDICATOR (AKU 2014 – 2025)

- ★ What are fact finding techniques?
- ★ Explain feasibility study and its types.
- ★ What is cost-benefit analysis?
- ★ What is prototyping? Explain its types.
- ★ What is JAD? Explain its benefits.
- ★ Why is requirement gathering important?



SHORT TRICK BOX

FACT FINDING: I-G-S-P-Q
(Interview, Group, Site visit, Presentation, Questionnaire)

FEASIBILITY: T-O-E-L
(Technical, Operational, Economic, Legal)

JAD: U-A-M (Users, Analysts, Managers)



2-MARK READY ANSWER (ANY TOPIC)

- Fact finding = collecting information about current system.
- Feasibility study checks practicality and benefits.
- Cost-benefit analysis compares cost and benefits.
- JAD involves users, analysts and managers together.
- Prototyping is sample model for user feedback.

UNIT-5 FINAL ONE LINE

Good planning, complete information, user involvement, and proper feasibility study lead to successful system development.





BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-6 : MODULAR & STRUCTURED DESIGN + SYSTEM DESIGN & MODELLING



1. DESIGN PRINCIPLES

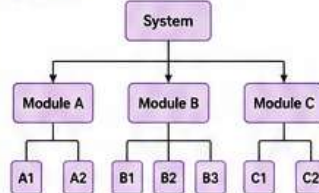
- Design is the process of converting requirements into a blueprint for building the system.
- Good design leads to a system that is reliable, efficient and easy to maintain.
- Principles of good design:
 - ✓ Keep it simple (Simplicity)
 - ✓ Divide and conquer (Modularity)
 - ✓ High cohesion, low coupling
 - ✓ Information hiding
 - ✓ Reusability
 - ✓ Flexibility
 - ✓ Efficiency



2. TOP DOWN DESIGN

- Start from the whole system and break it down into smaller modules.
- Each module is further divided until detail level is reached.
- Easy to manage and control.
- Used in early stages of design.

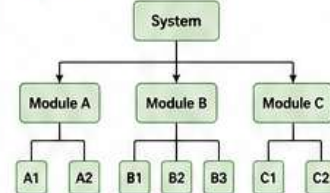
Diagram:



3. BOTTOM UP DESIGN

- Start from small modules and combine them to form larger modules.
- Focus on details first and then integrate.
- Useful when modules are reusable.
- Ensures reliability at lower level.

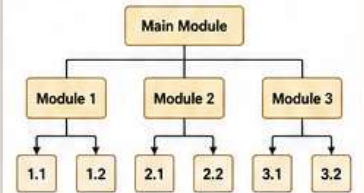
Diagram:



4. STRUCTURE CHARTS

- Graphical representation of modules and their relationships.
- Shows hierarchy and calling relationships.
- Helps in understanding system structure.

Structure Chart Example:



5. MODULARITY

- Dividing system into independent modules.
- Each module performs a specific task.

Benefits:

- ✓ Easy design and development
- ✓ Easy testing
- ✓ Easy maintenance
- ✓ Reusability



6. GOALS OF DESIGN

- 1 Correctness – Meets requirements.
- 2 Reliability – Works without failure.
- 3 Efficiency – Uses resources optimally.
- 4 Usability – Easy to use.
- 5 Maintainability – Easy to modify.
- 6 Flexibility – Easy to adapt to changes.



7. COUPLING

- Coupling is the degree of interdependence between modules.

Types of Coupling (Low to High):

- 1 Data Coupling – Communicate through data.
- 2 Stamp Coupling – Pass composite data.
- 3 Control Coupling – Pass control information.
- 4 External Coupling – Depend on external device/file format.
- 5 Common Coupling – Share global data.
- 6 Content Coupling – One module uses the internal data of another module.

Good Design: Low Coupling

8. COHESION

- Cohesion is the degree to which elements inside a module belong together.

Types of Cohesion (Low to High):

- 1 Coincidental Cohesion – No relation.
- 2 Logical Cohesion – Grouped by category.
- 3 Temporal Cohesion – Performed at same time.
- 4 Procedural Cohesion – Follow sequence.
- 5 Communicational Cohesion – Operate on same data.
- 6 Sequential Cohesion – Output used as input to next.
- 7 Functional Cohesion – Single well-defined function.

Good Design: High Cohesion

9. LOGICAL AND PHYSICAL DESIGN

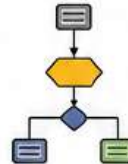
Basis	Logical Design	Physical Design
Meaning	Focuses on what the system will do.	Focuses on how the system will be implemented.
Focus	Independent of hardware and software.	Dependent on hardware, software and technology.
Output	DFD, ER Diagram, Data Dictionary, Process Specs.	Database design, file structures, program design, interface design.
Purpose	To create logical model of the system.	To convert logical model into physical implementation.
Level	High level.	Low level.
Example	What data to store, what process to perform.	How data will be stored, what DBMS will be used.
Change Impact	Easy to change.	Changes are costly.

10. PROCESS MODELLING

It represents how the system processes work.

Tools:

- ✓ Data Flow Diagrams (DFD)
- ✓ Flowcharts
- ✓ Structure Charts
- ✓ Decision Tables
- ✓ Decision Trees



11. DATA FLOW DIAGRAMS (DFD)

DFD shows flow of data in the system.

Symbols Used:

	External Entity	Source or destination of data
	Process	Transforms input into output
	Data Flow	Flow of data
	Data Store	Stores data
	Control Flow	Sequence/control information

Levels of DFD:

- Level 0 (Context Diagram) – Whole system as single process.
- Level 1 – Major processes.
- Level 2 – Sub-processes of Level 1.



12. DATA MODELLING

It is the process of defining data, data relationships, semantics and constraints.

Purpose:

- ✓ To design database.
- ✓ To organize data.

Tools:

- ER Diagram
- Data Dictionary



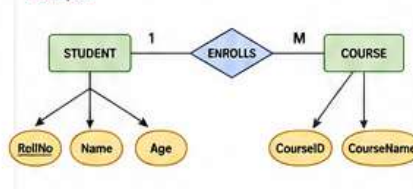
13. E-R DIAGRAM (ENTITY-RELATIONSHIP DIAGRAM)

Graphical representation of entities, attributes and relationships.

Components:

- Entity: A real world object
- Relationship: Association between entities
- Attribute: Property of an entity
- Weak Entity: Depends on another entity

Example:



14. PROCESS SPECIFICATION TOOLS

Tools used to specify how each process works.

Types:

- Structured English
- Decision Tables
- Decision Trees
- Flowcharts
- Pseudo Code



15. DECISION TABLE

Table showing conditions and corresponding actions.

Example:

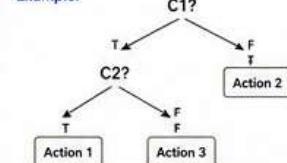
Conditions		Actions	
C1	C2	Action 1	Action 2
T	T	X	
T	F		X
F	T		
F	F		X

T = True, F = False, X = Action to be taken

16. DECISION TREE

Tree structure showing conditions and outcomes.

Example:



17. STRUCTURED ENGLISH

English-like language using structured control statements.

Example:

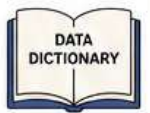
```
IF marks >= 40 THEN
  PRINT "Pass"
ELSE
  PRINT "Fail"
END IF
```

18. DATA DICTIONARY

Repository of information about data used in the system.

Contents:

- ✓ Data element name
- ✓ Description
- ✓ Alias
- ✓ Data type
- ✓ Size
- ✓ Valid values
- ✓ Source
- ✓ Usage



PYQ INDICATOR (AKU 2014 - 2025)

- ★ Explain Top-Down Design.
- ★ Explain Bottom-Up Design.
- ★ What is Modularity? Write its benefits.
- ★ Differentiate Coupling and Cohesion.
- ★ What is DFD? Draw symbols.
- ★ What is ER Diagram? Draw example.
- ★ Explain Decision Table and Decision Tree.



SHORT TRICK BOX

- Low Coupling = Good Design
- High Cohesion = Good Design
- Top-Down = Break Down
- Bottom-Up = Build Up
- DFD Levels = 0, 1, 2
- Decision Table = Conditions + Actions
- Decision Tree = Tree of Decisions



2-MARK READY ANSWER (ANY TOPIC)

- Top-Down Design starts from main system and breaks it.
- Bottom-Up Design starts from small modules and combines.
- Coupling is interdependence between modules.
- Cohesion is relatedness within a module.
- DFD is used to show data flow in a system.
- ER Diagram shows entities, attributes and relationships.
- Decision Table lists conditions and actions in table.



ONE LINE SUMMARY

Good design uses modular approach, proper modelling (DFD, ER), high cohesion, low coupling and suitable tools for effective and maintainable system implementation.



UNIT-6 FINAL ONE LINE: Good design with proper modelling and structured approach results in a reliable, efficient and easy to maintain system.



BCA-203 SYSTEM ANALYSIS & DESIGN

UNIT-7 : MORE DESIGN ISSUES, CASE TOOLS, FORMS & REPORTS

DESIGN + AUDIT AND SECURITY OF COMPUTER SYSTEMS



1. MORE DESIGN ISSUES AND CASE TOOLS

More Design Issues

- User involvement
- Change management
- Data security and integrity
- System reliability
- Performance and scalability
- Flexibility and adaptability
- Usability and simplicity
- Documentation and training
- Cost and time constraints
- Standards and compliance



CASE Tools (Computer Aided Software Engineering)

CASE tools support analysts and designers in developing and maintaining systems.

Benefits:

- ✓ Increases productivity
- ✓ Ensures consistency
- ✓ Improves quality
- ✓ Reduces development time
- ✓ Better documentation and maintenance

2. FORMS AND REPORTS DESIGN

Forms

Forms are used for collecting data from users.



Reports

Reports are used for presenting processed information to users.



Differences between Forms and Reports

Basis	Forms	Reports
Purpose	Input / Data Collection	Output / Information Display
Direction	Filled by user	Read by user
Frequency	Many times	Less frequent
Design	Simple and structured	More complex, summarized
Example	Application form, Survey form	Sales report, Student report

3. PROCESS OF DESIGNING FORMS AND REPORTS

- 1 Identify Purpose and Users
- 2 Gather Requirements
- 3 Plan the Content
- 4 Design Layout and Structure
- 5 Review and Validate
- 6 Test and Get Feedback
- 7 Finalize and Implement



Deliverables / Outcomes

- Well designed forms and reports
- Better data quality
- Improved decision making
- User satisfaction
- Time and cost savings

4. DESIGN SPECIFICATIONS AND NARRATIVE OVERVIEWS

Design Specifications

Detailed description of how the system will be designed.

Includes:

- System architecture
- Data design
- Process design
- Input/Output design
- File and database design
- Interface design
- Security design

Narrative Overviews

Description of system in simple language for users and management.

Includes:

- System purpose
- Major features
- How system will work
- Benefits
- Constraints



5. SAMPLE DESIGN AND TESTING & USABILITY ASSESSMENT

Sample Design

Creating sample forms and reports before final development.

Helps to:

- Visualize output
- Find errors early
- Get user feedback



Testing and Usability Assessment

Checking forms/reports for:

- Accuracy
- Completeness
- Readability
- Ease of use
- Navigation
- User satisfaction



6. TYPES OF INFORMATION

Internal Information

Information generated within the organization.

Examples:

- Employee details
- Sales records
- Inventory data
- Accounts data

External Information

Information received from outside the organization.

Examples:

- Market data
- Customer data
- Government rules
- Competitor data

Turnaround Document

Documents sent out to get a response and returned back with information.

Example: Order form, Feedback form.



7. GENERAL FORMATTING GUIDELINES FOR FORMS AND REPORTS

Meaningful Titles

Use clear and meaningful titles.



Meaningful Information

Provide only useful and necessary information.



Balanced Layout

Distribute content properly on the page.



Highlight Information

Use bold, italics, underlining to highlight important data.



Using Colour

Use limited colours consistently for better understanding.



Displaying Text

Use simple fonts and proper font size.



Easy Navigation

Make it easy to move from one section to another.



Guidelines for Displaying Contents

Use headings, subheadings, numbering and bullets.



Designing Tables and Lists

Use tables and lists for organized data.



Criteria for Form Design

Organization, Consistency, Completeness, Flexible Entry, Economy.



Criteria for Report Design



Relevance – Information should be relevant.



Clarity – Information be clear.



Timeliness – Information should be up-to-date.



Cost – Design should be cost effective.



8. AUDIT AND SECURITY OF COMPUTER SYSTEMS

Introduction

Audit and security ensure that computer systems work properly, data is safe and resources are used effectively.



Definition of Audit

Audit is a systematic process of evaluating computer systems, operations and records to ensure accuracy, efficiency and compliance with policies.



Objectives of Audit

- Ensure accuracy and reliability of data
- Check compliance with policies and procedures
- Prevent and detect fraud and errors
- Evaluate efficiency of operations
- Safeguard assets and resources
- Provide recommendations for improvement



10. SECURITY OF COMPUTER SYSTEMS

Security Goals (CIA Triad)



Confidentiality
Information should be accessible only to authorized users.



Integrity
Information should be accurate and not be changed without authorization.



Availability
Information and resources should be available when needed.

Security Threats

- Unauthorized access
- Malware and viruses
- Hacking
- Data theft
- Hardware failure
- Natural disasters

Security Measures

- Access control
- Strong passwords
- Encryption
- Firewalls
- Antivirus software
- Regular backup
- Audit trails

9. TYPES OF AUDIT

System Audit

Evaluates overall system performance, controls and procedures.



Operational Audit

Checks day-to-day operations and ensures efficiency.



Compliance Audit

Ensures rules, regulations and policies are followed.



Information Audit

Evaluates accuracy, quality and security of information.



11. ONE LINE SUMMARY

- ✓ Forms collect data; Reports present information.
- ✓ Good design improves usability and decision making.
- ✓ Internal information comes from within the organization.
- ✓ External information comes from outside sources.
- ✓ Turnaround documents are sent and returned for response.
- ✓ Use clear titles, simple layout and proper formatting.
- ✓ Sample design and testing ensure quality output.
- ✓ CASE tools improve productivity and consistency.
- ✓ Audit ensures accuracy, compliance and efficiency.
- ✓ Security protects systems from threats and unauthorized access.



PYQ INDICATOR (AKU 2014 - 2025)

- ★ Explain different types of documentation.
- ★ What are the guidelines for form design?
- ★ Difference between forms and reports.
- ★ What is audit? Write its objectives.
- ★ Explain data dictionary.
- ★ Discuss decision table and decision tree.



SHORT TRICK BOX

- ★ Forms vs Reports → F = Fill, R = Read
- ★ CIA → Confidentiality, Integrity, Availability
- ★ Good Form Design → O.C.C.F.E (Organization, Consistency, Completeness, Flexible Entry, Economy)
- ★ Audit Objectives → A.C.P.E.S.P (Accuracy, Compliance, Prevention, Efficiency, Safeguard, Provide Improvement)



2-MARK READY ANSWER (ANY TOPIC)

- Forms are for input; reports are for output.
- Good documentation ensures clarity and quality.
- CASE tools help in analysis, design and maintenance.
- Audit checks systems for accuracy and compliance.
- CIA triad ensures security goals.



UNIT-7 FINAL ONE LINE

Well designed forms, reports, proper documentation, secure systems and regular audit lead to effective and reliable information systems.



UNIT-7 FINAL ONE LINE: Good design, proper documentation, security and audit ensure reliable, efficient and quality systems.





BCA-203 SYSTEM ANALYSIS & DESIGN

FINAL REVISION CHEAT SHEET (ALL 7 UNITS)



UNIT 1: INTRODUCTION TO SAD

Key Points

- SAD converts user requirements into a blueprint for system.
- Good design → Efficient, reliable, easy to maintain system.
- Principles: Simplicity, Modularity, High cohesion, Low coupling, Information hiding, Reusability, Flexibility, Efficiency.

Objectives

- Meet user needs effectively
- Use resources optimally
- High quality & reliable system
- Easy maintenance & adaptability

SAD Life Cycle



Types of Systems

- Manual System – No computer use
- Automated System – Some automation
- Information System – Process data
- Decision Support System – Helps decision
- Expert System – Uses AI & knowledge base

UNIT 2: SYSTEMS ANALYST – A PROFESSION

Role of Systems Analyst

- Study current system
- Identify problems & opportunities
- Analyze requirements
- Design and evaluate solutions
- Supervise development
- Ensure system meets user needs

Skills Required

- Technical Knowledge
- Analytical Ability
- Communication Skills
- Problem Solving
- Organization & Management



Responsibilities

- Gather and analyze requirements
- Feasibility study
- Prepare specification documents
- Coordinate with users & developers
- Testing and evaluation
- Ensure proper implementation

Ethics of Analyst

- Honesty and integrity
- Confidentiality
- Competence and diligence
- Respect for users' interest
- Avoid conflicts of interest

Qualities

- ✓ Good Listener ✓ Impartial ✓ Patient ✓ Creative ✓ Detail Oriented ✓ Team Player

UNIT 3: PROCESS OF SYSTEM DEVELOPMENT

SDLC Phases

- 1 Project Identification & Selection
- 2 Project Initiation & Planning
- 3 Analysis
- 4 Logical Design
- 5 Physical Design
- 6 Implementation
- 7 Maintenance

SDLC Models

- Waterfall Model – Linear & sequential
- Prototyping Model – Build prototype, get feedback
- Incremental Model – Build in parts
- Spiral Model – Risk-driven, iterative
- Agile Model – Iterative & flexible

Feasibility Study

Checks practicality of system development.
Types: Technical | Operational | Economic | Legal
A feasible system is: Practical, Acceptable, Affordable, Legal

FACT FINDING TECHNIQUES



UNIT 4: DOCUMENTATION OF SYSTEMS

Types of Documentation

1. Planning Documents – Plans, schedules, budget
2. Requirement Documents – User & system needs
3. Design Documents – Logical & physical design
4. Development Documents – Code, program logic
5. Testing Documents – Test cases, results
6. Operation Documents – User manual, procedures
7. Maintenance Documents – Change records

System Requirement Specification (SRS)

Describes what the system will do.

Contents:

- Introduction
- Overall description
- Functional requirements
- Non-functional requirements
- External interfaces
- Constraints
- Acceptance criteria

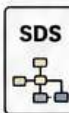


Design Specification (SDS)

Describes how the system will meet requirements.

Contents:

- System architecture
- Data design
- Process design
- Input/Output design
- File & database design
- Security & interface design



User Manual

Helps users understand & use the system.

Includes:

- Introduction
- System overview
- Installation
- Usage instructions
- Examples
- Troubleshooting
- FAQ



Test Design Document

Describes testing approach, test cases & expected results.

Includes:

- Test objectives
- Test strategy
- Test cases
- Test data
- Expected results



Good Documentation ✓ Clear ✓ Accurate ✓ Complete ✓ Consistent ✓ Up-to-date ✓ Easy to understand

UNIT 5: PLANNING AND DESIGNING SYSTEMS

System Planning Steps

- 1 Identify Problems / Opportunities
- 2 Set Objectives & Goals
- 3 Study Current System
- 4 Gather Information
- 5 Conduct Feasibility Study
- 6 Select Best Solution
- 7 Prepare Plan & Schedule
- 8 Get Approval & Start Project

Cost-Benefit Analysis

Compare total cost vs total benefits.

Costs	Benefits
• Hardware	• Increase in profit
• Software	• Better service
• Development	• Time saving
• Training	• Accuracy
• Maintenance	• Better control
• Operating	• Competitive advantage

Tools for Planning

- Gantt Chart – Project scheduling
- PERT Chart – Time & cost estimation
- Network Diagram – Activity flow

Gantt Chart Example

Activity	Week 1	Week 2	Week 3	Week 4
Analysis				
Design				
Development				
Testing				
Implementation				

Requirement Gathering

- Functional Requirements – What system should do
- Non-functional Requirements – How well it should do

Sources: Users, Managers, Existing documents, Observation, Surveys, Questionnaires

UNIT 6: MODULAR & STRUCTURED DESIGN + SYSTEM DESIGN & MODELLING

Modular Design

- Divides system into independent modules.
- Each module performs a specific task.
- Benefits: Easy design, testing, maintenance, reusability.



Top-Down Design

- Start from overall system & break into smaller modules.
- Refinement until detail level reached.
- Easy to manage & control.



Bottom-Up Design

- Start from small modules & combine to form larger system.
- Focus on details first.
- Ensures reusability & reliability.



Structured Design Concepts

Coupling – Interdependence between modules

Types (Low to High):
Data, Stamp, Control, External, Common, Content Coupling



Cohesion – Relatedness within module

Types (Low to High):
Coincidental, Logical, Temporal, Procedural, Communicational, Sequential, Functional

Design Modelling Tools

- Data Flow Diagrams (DFD)
- Flowcharts
- Structure Charts
- Decision Tables
- Decision Trees
- Pseudo Code



Data Modelling

- ER Diagram
- Data Dictionary



Key Principles

- ✓ High Cohesion ✓ Low Coupling ✓ Information Hiding ✓ Modularity ✓ Abstraction

UNIT 7: MORE DESIGN ISSUES, CASE TOOLS, FORMS & REPORTS DESIGN + AUDIT AND SECURITY OF COMPUTER SYSTEMS

More Design Issues

- User involvement
- Change management
- Data security & integrity
- System reliability
- Performance & scalability
- Flexibility & adaptability
- Usability & simplicity
- Documentation & training
- Cost & time constraints
- Standards & compliances

CASE Tools (Computer Aided Software Engineering)

Tools to support analysts and designers.

Benefits:

- Increases productivity
- Ensures consistency
- Improves quality
- Reduces development time
- Better documentation & maintenance



Forms Design Guidelines

- Clear title & purpose
- Simple & easy to fill
- Proper layout & structure
- Use proper labels
- Avoid unnecessary fields
- Provide instructions
- Ensure accuracy & readability



Reports Design Guidelines

- Clear title & summary
- Relevant & accurate data
- Proper headings & subheadings
- Easy to read & understand
- Use tables, charts & graphs
- Consistent format
- Focus on user needs



Audit of Computer Systems

Types:

- System Audit – Overall system performance & controls
- Operational Audit – Day to day operations
- Compliance Audit – Policies & regulations



Security of Computer Systems

Goals (CIA Triad):

- Confidentiality
- Integrity
- Availability

Measures:

- Access control
- Strong passwords
- Encryption
- Firewalls
- Antivirus
- Regular backups
- Audit trails



IMPORTANT POINTS

- ★ Good planning → Successful system.
- ★ Good design → Quality system.
- ★ Documentation → Communication & future reference.
- ★ Testing → Quality assurance.
- ★ Maintenance → Long term success.



SHORT TRICK BOX

SDLC Phases: P-A-D-I-M
CASE Benefits: P-C-Q-D-D
Feasibility Types: T-O-E-L
Cohesion (Low→High):
C-L-T-P-C-S-F



PYQ INDICATOR (AKU 2014 – 2025)

- ✓ Explain SDLC phases.
- ✓ What is feasibility study? Types?
- ✓ Differentiate Top-Down & Bottom-Up Design.
- ✓ Draw DFD symbols.
- ✓ Explain different types of documentation.
- ✓ What are design principles?



ONE LINE SUMMARY

System Analysis & Design is the process of understanding user needs, analyzing requirements, designing, documenting, implementing and maintaining systems to meet goals efficiently and securely.



UNIT 1

INTRODUCTION TO SAD

Meaning

SAD is the process of studying a system to improve it and design a new system.

Objectives

- Meet user needs
- Efficient & reliable system
- Optimum use of resources
- Easy maintenance

Principles

- Simplicity
- Modularity
- High cohesion, Low coupling
- Information hiding
- Reusability, Flexibility, Efficiency

SAD Life Cycle

- 1 Planning
- 2 Analysis
- 3 Design
- 4 Implementation
- 5 Maintenance

Types of Systems

- Manual System
- Automated System
- Information System
- Decision Support System
- Expert System



UNIT 2

SYSTEMS ANALYST – A PROFESSION

Role of Systems Analyst

- Study current system
- Identify problems & opportunities
- Analyze requirements
- Design & evaluate solutions
- Feasibility study
- Supervise development
- Ensure system meets user needs

Skills Required

- Technical Knowledge
- Analytical Ability
- Communication Skills
- Problem Solving
- Organization & Management

Responsibilities

- Gather & analyze requirements
- Prepare SRS
- Coordinate users & developers
- Testing & evaluation
- Ensure proper implementation

Qualities

- ✓ Good Listener
- ✓ Patient
- ✓ Detail Oriented
- ✓ Impartial
- ✓ Creative
- ✓ Team Player

Ethics of Analyst

- Honesty & integrity
- Confidentiality
- Competence & diligence
- Respect for users' interest
- Avoid conflicts of interest



UNIT 3

PROCESS OF SYSTEM DEVELOPMENT

SDLC Phases

- 1 Project Identification & Selection
- 2 Project Initiation & Planning
- 3 Analysis
- 4 Logical Design
- 5 Physical Design
- 6 Implementation
- 7 Maintenance



SDLC Models

- Waterfall Model
- Prototyping Model
- Incremental Model
- Spiral Model
- Agile Model



Fact Finding Techniques

- Interviews
- Questionnaires
- Observation
- Sampling of Documents
- Group Discussion
- Site Visits
- Presentations

UNIT 5

PLANNING AND DESIGNING SYSTEMS

System Planning Steps

- 1 Identify Problems/Opportunities
- 2 Set Objectives & Goals
- 3 Study Current System
- 4 Gather Information
- 5 Conduct Feasibility Study
- 6 Select Best Solution
- 7 Prepare Plan & Schedule
- 8 Get Approval & Start Project



Tools for Planning

- Gantt Chart – Project scheduling
- PERT Chart – Time & cost estimation
- Network Diagram – Activity flow

Gantt Chart Example

Activity	Week 1	Week 2	Week 3	Week 4
Analysis	■			
Design		■		
Development			■	
Testing				■
Implementation				■

Cost-Benefit Analysis

Compare total cost vs total benefits.

- | | |
|--|--|
| Costs <ul style="list-style-type: none"> • Hardware • Software • Development • Training • Maintenance • Operating | Benefits <ul style="list-style-type: none"> • Increase in profit • Better service • Time saving • Accuracy • Better control • Competitive advantage |
|--|--|



Requirement Gathering

- Functional Requirements – What system should do
- Non-functional Requirements – How well it should do

Sources: Users, Managers, Existing documents, Observation, Surveys, Questionnaires



UNIT 4

DOCUMENTATION OF SYSTEMS

Types of Documentation

1. Planning Documents
2. Requirement Documents
3. Design Documents
4. Development Documents
5. Testing Documents
6. Operation Documents
7. Maintenance Documents



SRS (System Requirement Specification)

Describes what the system will do.

Contents:

- Introduction
- Overall description
- Functional requirements
- Non-functional requirements
- External interfaces
- Constraints
- Acceptance criteria



SDS (System Design Specification)

Describes how the system will meet requirements.

- System architecture
- Data design
- Process design
- I/O design
- File & database design
- Security & interface design



User Manual

Helps users understand & use the system.

Includes:

- Introduction
- System overview
- Installation
- Usage instructions
- Examples
- Troubleshooting



Test Design Document

Describes testing approach, test cases & expected results.

Includes:

- Test objectives
- Test strategy
- Test cases
- Test data
- Expected results



UNIT 6

MODULAR & STRUCTURED DESIGN + SYSTEM DESIGN & MODELLING

Modular Design

- Divide system into independent modules
- Each module performs a specific task
- Benefits: Easy design, testing, maintenance, reusability



Top-Down Design

- Start from overall system & break into smaller modules
- Refine until detail level reached
- Easy to manage & control



Bottom-Up Design

- Start from small modules & combine to form larger system
- Focus on details first
- Ensures reusability & reliability



Structured Design Concepts

Coupling (Interdependence)

Types (Low to High): Data, Stamp, Control, External, Common, Content (Prefer Low Coupling)



Cohesion (Relatedness)

Types (Low to High): Coincidental, Logical, Temporal, Procedural, Communicational, Sequential, Functional (Prefer High Cohesion)



Design Modelling Tools

- DFD (Data Flow Diagram)
- Flowcharts
- Structure Charts
- Decision Tables
- Decision Trees
- Pseudo Code



Data Modelling

- ER Diagram
- Data Dictionary



UNIT 7

MORE DESIGN ISSUES, CASE TOOLS, FORMS & REPORTS DESIGN + AUDIT & SECURITY OF COMPUTER SYSTEMS

More Design Issues

- User involvement
- Change management
- Data security & integrity
- System reliability
- Performance & scalability
- Flexibility & adaptability
- Usability & simplicity
- Documentation & training
- Cost & time constraints
- Standards & compliance

CASE Tools

Computer Aided Software Engineering tools help in:

- Increasing productivity
- Ensuring consistency
- Improving quality
- Reducing development time
- Better documentation & maintenance



Forms Design Guidelines

- Clear title & purpose
- Simple & easy to fill
- Proper layout & structure
- Use proper labels
- Avoid unnecessary fields
- Provide instructions
- Ensure accuracy & readability



Reports Design Guidelines

- Clear title & summary
- Relevant & accurate data
- Proper headings & subheadings
- Easy to read & understand
- Use tables, charts & graphs
- Consistent format
- Focus on user needs



Audit of Computer Systems

- Types:
- System Audit – Overall system contranacs
- Operational Audit – Day to day operations
- Compliance Audit – Policies & Reg regulations



Security of Computer Systems

- CIA Triad Goals:
- Confidentiality – Info for authorized users only
- Integrity – Accurate & not altered
- Availability – Info & resources when needed
- Security Measures:
- Access Control, Strong Passwords, Encryption, Firewalls, Antivirus, Regular Backups, Audit Trails



KEY CONCEPTS SUMMARY

- ✓ SAD is a problem-solving approach to develop better information systems.
- ✓ SDLC provides a structured process to build high-quality systems.
- ✓ Analyst plays a vital role in understanding & fulfilling user needs.
- ✓ Documentation ensures clarity, communication & future reference.
- ✓ Good planning & design ensure efficiency, reliability & effectiveness.
- ✓ Modular & structured design improve manageability & reusability.
- ✓ CASE tools, good forms & reports improve productivity & quality.
- ✓ Audit & security protect systems & information from risks.



SUCCESS FORMULA





BCA-203 SYSTEM ANALYSIS & DESIGN

★★ MOST EXPECTED PYQs (2014–2025) CHEAT SHEET ★★



UNIT-1: INTRODUCTION TO SAD

- **System** – Set of interrelated components working together to achieve a goal.
- **Characteristics of System**
 - ✓ Organization ✓ Interaction ✓ Interdependence
 - ✓ Integration ✓ Central Objective
- **Types of System**
 - ✓ Physical / Abstract ✓ Natural / Man-made
 - ✓ Deterministic / Probabilistic
 - ✓ Open / Closed ✓ Simple / Complex
- **Real-time System** – Response must be within a specified time.
- **Distributed System** – Components at different locations connected by network.
- **Business Subsystems** – Support primary business functions.
- **SAD** – Process of studying a system and designing a better one.
- **Prototype Model** – Build a prototype, get feedback, refine & finalize.
- **JAD (Joint Application Development)** – Users, managers & analysts work together.



UNIT-2: SYSTEMS ANALYST – A PROFESSION

- **Who is a Systems Analyst?**
Professional who studies systems, understands user needs and designs effective solutions.



• Role of Systems Analyst

- ✓ Study current system
- ✓ Analyze requirements
- ✓ Feasibility Study
- ✓ Implement & follow-up
- ✓ Identify problems
- ✓ Design solutions
- ✓ Evaluate solutions

Responsibilities

- ✓ Gather & analyze requirements
- ✓ Prepare SRS / Documents
- ✓ Coordinate users & developers
- ✓ Test & evaluate system
- ✓ Ensure proper implementation

Skills Required

- ✓ Technical Knowledge
- ✓ Analytical Ability
- ✓ Communication Skills
- ✓ Problem Solving
- ✓ Management Ability

Qualities: Good listener, Impartial, Patient, Creative, Detail oriented, Team player

Ethics: Honesty, Confidentiality, Competence, Avoid conflict of interest

UNIT-3: PROCESS OF SYSTEM DEVELOPMENT

SDLC Phases

- 1 Project Identification & Selection
- 2 Project Initiation & Planning
- 3 Analysis
- 4 Logical Design
- 5 Physical Design
- 6 Implementation
- 7 Maintenance

SDLC Models

- Waterfall Model
- Prototyping Model
- Incremental Model
- Spiral Model
- Agile Model



SDLC Phase Details

1. Identification – Identify problems & opportunities
2. Planning – Define objectives, resources, schedule
3. Analysis – Study current system, gather requirements
4. Design – Logical & physical design
5. Implementation – Coding, testing, training, installation
6. Maintenance – Corrective, Adaptive, Perfective, Preventive

Other Approaches

- Prototyping • JAD • Participatory Design

UNIT-4: DOCUMENTATION OF SYSTEMS

Types of Documentation

1. Planning Documents
2. Requirement Documents
3. Design Documents
4. Development Documents
5. Testing Documents
6. Operation Documents
7. Maintenance Documents



SRS (System Requirement Specification)

Defines what the system will do.

Contents:

- Introduction
- Overall Description
- Functional Requirements
- Non-functional Requirements
- External Interfaces
- Constraints
- Acceptance Criteria



SDS (System Design Specification)

Defines how the system will meet requirements.

Includes:

- System Architecture
- Data Design
- Process Design
- I/O Design
- File & Database Design
- Security & Interface Design



User Manual

Helps users understand & use the system.

Includes:

- Introduction
- System Overview
- Installation
- Usage Instructions
- Examples
- Troubleshooting



Test Design Document

Describes testing approach, test cases & expected results.

Includes:

- Test Objectives
- Test Strategy
- Test Cases
- Test Data
- Expected Results



Good Documentation

- ✓ Clear ✓ Accurate ✓ Complete ✓ Consistent ✓ Up-to-date ✓ Easy to Understand

UNIT-5: PLANNING AND DESIGNING SYSTEMS

System Planning Steps

- 1 Identify Problems / Opportunities
- 2 Set Objectives & Goals
- 3 Study Current System
- 4 Gather Information
- 5 Conduct Feasibility Study
- 6 Select Best Solution
- 7 Prepare Plan & Schedule
- 8 Get Approval & Start Project

Fact Finding Techniques

- Interview (One-to-one talk)
- Questionnaire (Written set)
- Observation (Watch work)
- Sampling of Documents
- Group Discussion
- Site Visit (Observation)
- Presentations

Feasibility Study

- Technical** Is technology available?
- Operational** Will system work in organization?
- Economic** Is it cost effective?
- Legal** Is it legally acceptable?

Cost-Benefit Analysis

- | Costs | Benefits |
|--------------------|-------------------------|
| • Hardware Cost | • Increase in Profit |
| • Software Cost | • Better Service |
| • Development Cost | • Time Saving |
| • Training Cost | • Accuracy |
| • Operating Cost | • Better Control |
| • Maintenance Cost | • Competitive Advantage |



Requirement Gathering

- **Functional Requirements** – What system should do
- **Non-functional Requirements** – How well it should do

Sources: Users, Managers, Documents, Observation, Surveys

UNIT-6: MODULAR & STRUCTURED DESIGN + SYSTEM DESIGN & MODELLING

Modular Design Approaches

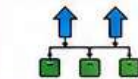
Top-Down Design

Start from overall system & break into modules.



Bottom-Up Design

Start from small modules & combine to form system.



Hybrid Approach

Combination of Top-Down and Bottom-Up.



Coupling (Interdependence)

Low Coupling (Good)
Modules are independent



High Coupling (Bad)
Modules highly dependent



Cohesion (Relatedness)

High Cohesion (Good)
Elements in module are closely related.



Low Cohesion (Bad)
Unrelated elements in a module.



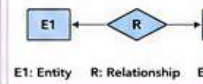
Design & Modelling Tools

Structured Design Tools

- Structure Chart
- Data Flow Diagram (DFD)
- Decision Table
- Decision Tree
- Structured English
- Data Dictionary

Data Modelling Tools

- ER Diagram
- Data Dictionary



DFD Levels

- Context Diagram (Level 0) – Whole system as one process
- Level 1 – Main processes
- Level 2 – Sub-processes of Level 1

UNIT-7: MORE DESIGN ISSUES, CASE TOOLS, FORMS & REPORTS DESIGN + AUDIT & SECURITY OF COMPUTER SYSTEMS

More Design Issues

- User Involvement
- Change Management
- Data Security & Integrity
- System Reliability
- Performance & Scalability
- Flexibility & Adaptability
- Usability & Simplicity
- Documentation & Training
- Cost & Time Constraints
- Standards & Compliance



CASE Tools (Computer Aided Software Engineering)

Tools to support analysts & designers in SDLC.

Benefits:

- ✓ Increase Productivity
- ✓ Ensure Consistency
- ✓ Improve Quality
- ✓ Reduce Development Time
- ✓ Better Documentation
- ✓ Easy Maintenance



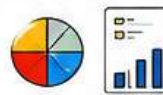
Forms Design Guidelines

- Clear Title & Purpose
- Simple & Easy to Fill
- Proper Layout & Structure
- Use Proper Labels
- Avoid Unnecessary Fields
- Provide Instructions
- Ensure Accuracy & Readability



Reports Design Guidelines

- Clear Title & Summary
- Relevant & Accurate Data
- Proper Headings & Subheadings
- Easy to Read & Understand
- Use Tables, Charts & Graphs
- Consistent Format
- Focus on User Needs



Audit of Computer Systems

Objectives

- ✓ Ensure Accuracy
- ✓ Ensure Security
- ✓ Ensure Efficiency
- ✓ Ensure Compliance
- ✓ Prevent & Detect Errors
- ✓ Evaluate Effectiveness



Security of Computer Systems

Goals (CIA Triad)

- Confidentiality
- Integrity
- Availability



Security Measures

- Access Control
- Strong Passwords
- Encryption
- Firewalls
- Antivirus
- Regular Backups
- Audit Trails



TOP 20 MOST REPEATED QUESTIONS

1. SDLC with diagram
2. Types of Systems
3. System Analyst
4. Role of System Analyst
5. SRS (System Requirement Specification)
6. DFD (Data Flow Diagram)
7. ER Diagram
8. Coupling & Cohesion
9. Top Down vs Bottom Up
10. Prototype Model
11. Feasibility Study
12. Cost Benefit Analysis
13. Documentation
14. CASE Tools
15. Forms vs Reports
16. Audit
17. Joint Application Development (JAD)
18. Real Time System
19. Distributed System
20. Logical vs Physical Design



SHORT TRICK BOX

- SDLC Phases: P-I-A-D-I-M (Plan-Identify-Analyze-Design-Implement-Maintain)
- Feasibility Types: T-O-E-L (Technical-Operational-Economic-Legal)
- CIA Triad: C-I-A (Confidentiality-Integrity-Availability)
- Coupling (Low is Good): D-S-C-E-X (Data-Stamp-Control-External-Content)
- Cohesion (High is Good): C-L-T-P-S-F (Coincidental-Logical-Temporal-Procedural-Sequential-Functional)



EXAM TIP

- ✓ Draw Diagrams wherever required.
- ✓ Use Headings & Subheadings.
- ✓ Write points in Bullet form.
- ✓ Explain with Examples.
- ✓ Diagrams of DFD, ER, Structure Chart carry more marks.
- ✓ Read Question carefully.
- ✓ Manage Time in Exam Well.



ONE LINE SUMMARY

- SAD helps develop better information systems.
- SDLC provides a structured development path.
- Good documentation ensures clarity & quality.
- Feasibility helps choose the right solution.
- Modular & structured design improves quality.
- CASE tools increase productivity.
- Forms & Reports present information effectively.
- Audit & Security protect system from risks.





BCA-203 SYSTEM ANALYSIS & DESIGN

MCQ + ONE-LINER REVISION SHEET (ALL 7 UNIT)



UNIT-1: INTRODUCTION TO SAD

MCQ

- SAD stands for?
a) System Approach & Design
b) System Analysis & Design
c) Structured Analysis & Design
d) Software Analysis & Design
- Which of the following is NOT a characteristic of a system?
a) Organization b) Interaction
c) Chaos d) Interdependence
- Real-time system must respond within?
a) Minutes b) Hours
c) Deadlines d) Days
- Which model builds a working prototype?
a) Waterfall b) Prototype
c) Spiral d) Incremental
- JAD stands for?
a) Joint Application Development
b) Joint Analysis & Design
c) Joint Advanced Development
d) Joint Automated Design

ONE-LINERS

- A system is a set of interrelated components working together.
- Characteristics: Organization, Interaction, Interdependence, Integration, Central Objective.
- Real-time systems respond within a fixed time.
- Distributed system works on multiple computers.
- Business subsystem supports main business functions.
- Prototype model is used to understand user requirements.
- JAD involves users, managers and analysts together.



UNIT-2: SYSTEMS ANALYST – A PROFESSION

MCQ

- Who is a Systems Analyst?
a) Programmer b) End User
c) Professional who studies systems d) Operator
- Main role of a Systems Analyst is?
a) Data entry b) System study
c) System maintenance d) Typing
- Which skill is NOT required for systems analyst?
a) Technical b) Analytical
c) Cooking d) Communication
- Feasibility study is done to check?
a) System coding b) System errors
c) System viability d) System speed
- Which quality is essential for analyst?
a) Patient b) Creative c) Impartial
d) All of the above

ONE-LINERS

- A systems analyst studies, designs and improves systems.
- Role: Identify problems, analyze requirements, design solutions, evaluate and implement.
- Responsibilities include SRS preparation, user coordination, testing and evaluation.
- Skills: Technical, Analytical, Communication, Problem solving, Management.
- Good analyst is: Listener, Impartial, Patient, Detail oriented and Team player.



UNIT-3: PROCESS OF SYSTEM DEVELOPMENT

MCQ

- SDLC phases are?
a) 3 b) 4 c) 5 d) 7
- Which phase comes after Analysis?
a) Design b) Planning
c) Implementation d) Maintenance
- Which model is linear & sequential?
a) Spiral b) Prototype
c) Waterfall d) Incremental
- What is the last phase of SDLC?
a) Testing b) Implementation
c) Maintenance d) Design
- Participatory design means?
a) Only developer works
b) Only user works
c) Users participate in design
d) No design

ONE-LINERS

- SDLC = Project I.D → Planning → Analysis → Design → Implementation → Maintenance.
- Logical design is technology independent.
- Physical design is technology dependent.
- Prototyping builds model to get user feedback.
- JAD is a workshop based approach.
- Maintenance corrects, adapts and improves the system.



UNIT-4: DOCUMENTATION OF SYSTEMS

MCQ

- Documentation means?
a) Writing code b) Keeping records
c) Drawing diagram d) Testing
- Which document defines user requirements?
a) SRS b) SDS
c) Test Plan d) User Manual
- SDS stands for?
a) System Design Specification
b) Software Design System
c) System Data Structure
d) Software Document Standard
- Which document describes testing approach?
a) SRS b) SDS c) Test Design Doc
d) User Manual
- Good documentation is?
a) Confusing b) Clear
c) Lengthy d) Irrelevant

ONE-LINERS

- Documentation means recording all system information.
- Types: Planning, Requirement, Design, Development, Testing, Operation, Maintenance.
- SRS defines what the system will do.
- SDS defines how the system will meet requirements.
- User Manual helps users to operate the system.
- Test Design Document lists test cases and expected results.
- Good documentation is clear, accurate, complete and up-to-date.



UNIT-5: PLANNING AND DESIGNING SYSTEMS

MCQ

- Fact finding technique for collecting data is?
a) Coding b) Interview
c) Compiling d) Linking
- Economic feasibility checks?
a) Technical resources
b) Cost & benefits
c) User training d) System errors
- CBA stands for?
a) Cost Based Analysis
b) Cost Benefit Analysis
c) Computer Based Analysis
d) Cost Budget Analysis
- Which technique is used to gather information from users?
a) Interview b) Site visit
c) Both a & b d) None
- What is the aim of planning?
a) Write code b) Run system
c) Achieve objectives with minimum resources d) Design DB

ONE-LINERS

- Planning decides what, how, when and by whom the system will be developed.
- Fact finding techniques: Interview, Questionnaire, Observation, Site visit, Sampling, Group discussion.
- Feasibility Study checks viability of system.
- Feasibility types: Technical, Operational, Economic, Legal.
- CBA compares total cost and total benefits.
- Good plan reduces risk and increases success chances.



UNIT-6: MODULAR & STRUCTURED DESIGN + SYSTEM DESIGN & MODELLING

MCQ

- Modular design means?
a) Big modules b) Small modules
c) No modules d) Random modules
- Coupling refers to?
a) Relation within module
b) Interdependence between modules
c) Database relation d) User relation
- Cohesion refers to?
a) Relation between modules
b) Relation within module
c) Hardware relation d) Data relation
- A DFD shows?
a) Data flow b) Control flow
c) Program code d) Output only
- ER diagram is used for?
a) Process modelling
b) Data modelling
c) Control modelling
d) Structure chart

ONE-LINERS

- Modular design divides system into independent modules.
- Top-Down: Start from whole system.
- Bottom-Up: Combine small modules.
- Coupling (Low is good).
- Cohesion (High is good).
- DFD shows movement of data.
- ER diagram shows entities and relationships.
- Data Dictionary stores data details.



UNIT-7: MORE DESIGN ISSUES, CASE TOOLS, FORMS & REPORTS DESIGN + AUDIT & SECURITY OF COMPUTER SYSTEMS

MORE DESIGN ISSUES

MCQ

- Which is a design issue?
a) User involvement
b) Black and white monitor
c) Keyboard
d) Printer
- Which ensures system reliability?
a) Change management
b) Fancy design
c) Colour selection
d) Screen saver
- Which is a constraint?
a) Time b) Cost
c) Both a & b d) None

ONE-LINERS

- User involvement improves acceptance.
- Change management handles system changes.
- Data security protects data from unauthorized access.
- Constraints are limitations like cost, time, resources.
- Standards ensure consistency and quality.



CASE TOOLS

MCQ

- CASE stands for?
a) Computer Aided Software Engineering
b) Computer Automated System Engineering
c) Computer Assisted System Engineering
d) None
- Benefit of CASE tools?
a) Increase cost
b) Reduce productivity
c) Increase productivity
d) None

ONE-LINERS

- CASE tools support analysts and designers.
- Benefits: Productivity increase, Consistency, Quality, Reduce dev. time, Better documentation.
- Examples: ERwin, BPwin, Rational Rose, Power Designer.



FORMS & REPORTS

MCQ

- Which is used to collect data from user?
a) Report b) Form
c) File d) Chart
- A report is used to?
a) Collect data
b) Present information
c) Store data d) Input data
- Which guideline for good form design?
a) Use small font
b) Use clear labels
c) Use unnecessary fields
d) Complex layout

ONE-LINERS

- Form collects data.
- Report presents processed information.
- Form: Simple, Clear, Proper layout, User friendly.
- Report: Accurate, Relevant, Easy to read, Consistent format.



AUDIT & SECURITY

MCQ

- Audit is?
a) Checking system performance
b) Finding faults only
c) Stealing data
d) Writing code
- Main objective of audit?
a) Ensure accuracy
b) Increase speed
c) Reduce cost
d) All of the above
- Which protects system from unauthorized access?
a) Firewall b) Audit
c) Antivirus d) Backup

ONE-LINERS

- Audit evaluates system for accuracy, efficiency and security.
- Types: System audit, Operational audit, Compliance audit.
- Security goals (CIA): Confidentiality, Integrity, Availability.
- Security tools: Firewall, Antivirus, Encryption, Backup, Access control.



TOP REPEATED TOPICS (2014-2025)

- ★ SDLC & its phases
- ★ Types of Systems
- ★ System Analyst & its role
- ★ Feasibility Study
- ★ Fact Finding Techniques
- ★ Cost Benefit Analysis
- ★ SRS & its contents
- ★ Documentation types
- ★ Top-Down & Bottom-Up Design
- ★ Coupling & Cohesion
- ★ DFD & its levels
- ★ ER Diagram
- ★ Data Dictionary
- ★ Forms vs Reports
- ★ CASE Tools & Benefits
- ★ Audit – types & objectives
- ★ Security – CIA Triad
- ★ Logical vs Physical Design
- ★ Prototype Model
- ★ JAD (Joint Application Dev.)

ABBREVIATIONS / IMPORTANT TERMS

- SDLC : System Development Life Cycle
- SRS : System Requirement Specification
- SDS : System Design Specification
- DFD : Data Flow Diagram
- ERD : Entity Relationship Diagram
- CBA : Cost Benefit Analysis
- JAD : Joint Application Development
- CIA : Confidentiality, Integrity, Availability
- CASE : Computer Aided Software Engineering
- ROI : Return on Investment
- DBMS : Database Management System
- UML : Unified Modeling Language
- GUI : Graphical User Interface
- CRUD : Create, Read, Update, Delete
- API : Application Programming Interface
- KPI : Key Performance Indicator



ONE LINE SUMMARY (ALL UNITS)

- ✓ SAD improves systems by better analysis & design.
- ✓ SDLC provides a structured development path.
- ✓ Analyst studies, designs & implements systems.
- ✓ Documentation ensures clarity and future reference.
- ✓ Planning & feasibility ensure project success.
- ✓ Modular & structured design improves quality.
- ✓ CASE tools, forms, reports, audit & security ensure productivity, information quality & safety.





BCA-203 SYSTEM ANALYSIS & DESIGN

★ LAST-NIGHT REVISION POSTER ★

ALL 7 UNITS – HIGH WEIGHTAGE TOPICS – ONE SHEET REVISION



REVISION
TODAY
SUCCESS
TOMORROW

1 INTRODUCTION TO SAD

KEY CONCEPTS

- ★ SAD is the process of studying a system to improve it and design a new system.
- ★ Focuses on understanding user needs and developing better information systems.

CHARACTERISTICS OF SYSTEM

- Organization
- Interaction
- Interdependence
- Integration
- Central Objective

TYPES OF SYSTEM



OTHER IMPORTANT TOPICS

- Real-time System – Response within a fixed time.
- Distributed System – Components in different locations connected by network.
- Business Subsystems – Support primary business functions.
- Prototype Model – Build prototype → Get feedback → Refine.
- JAD – Users, Managers & Analysts work together.



2 SYSTEMS ANALYST – A PROFESSION

ROLE OF SYSTEMS ANALYST

- ✓ Study current system
- ✓ Identify problems & opportunities
- ✓ Analyze requirements
- ✓ Design & evaluate solutions
- ✓ Feasibility study
- ✓ Supervise development
- ✓ Ensure system meets user needs



RESPONSIBILITIES

- Gather & analyze requirements
- Prepare SRS / Documents
- Coordinate users & developers
- Testing & evaluation
- Ensure proper implementation

SKILLS REQUIRED

- Technical Knowledge
- Analytical Ability
- Communication Skills
- Problem Solving
- Management Ability

QUALITIES

Good Listener ★ Impartial ★ Patient ★ Creative
Detail Oriented ★ Team Player

ETHICS OF ANALYST

- Honesty & Integrity
- Confidentiality
- Competence & Diligence
- Respect for users' interest
- Avoid conflicts of interest

3 PROCESS OF SYSTEM DEVELOPMENT

SDLC PHASES

- 1 Project Identification & Selection
- 2 Project Initiation & Planning
- 3 Analysis
- 4 Logical Design
- 5 Physical Design
- 6 Implementation
- 7 Maintenance

SDLC MODELS

- Waterfall Model
- Prototyping Model
- Incremental Model
- Spiral Model
- Agile Model



OTHER APPROACHES

- Prototyping
- JAD
- Participatory Design

PHASE DETAILS (IN BRIEF)

- Analysis – Study current system & gather requirements.
- Logical Design – High level design (what the system will do).
- Physical Design – Technical details (how the system will do).
- Implementation – Coding, Testing, Training, Installation.
- Maintenance – Corrective, Adaptive, Perfective, Preventive.



4 DOCUMENTATION OF SYSTEMS

TYPES OF DOCUMENTATION

1. Planning Documents
2. Requirement Documents
3. Design Documents
4. Development Documents
5. Testing Documents
6. Operation Documents
7. Maintenance Documents

SRS (SYSTEM REQUIREMENT SPEC.)

Describes what the system will do.

CONTENTS

- Introduction
- Overall Description
- Functional Requirements
- Non-Functional Requirements
- External Interfaces
- Constraints
- Acceptance Criteria



SDS (SYSTEM DESIGN SPEC.)

Describes how the system will meet requirements.

- Includes:
- System Architecture
 - Data Design
 - Process Design
 - I/O Design
 - File & Database
 - Security & Interface Design



USER MANUAL

Helps users understand & use the system.

- Includes:
- Introduction
 - System Overview
 - Installation
 - Usage Instructions
 - Examples
 - Troubleshooting



TEST DESIGN DOCUMENT

Describes testing approach, test cases & expected results.

- Includes:
- Test Objectives
 - Test Strategy
 - Test Cases
 - Test Data
 - Expected Results



GOOD DOCUMENTATION IS: Clear • Accurate • Complete • Consistent • Up-to-date

5 PLANNING AND DESIGNING SYSTEMS

SYSTEM PLANNING STEPS

- 1 Identify Problems / Opportunities
- 2 Set Objectives & Goals
- 3 Study Current System
- 4 Gather Information
- 5 Conduct Feasibility Study
- 6 Select Best Solution
- 7 Prepare Plan & Schedule
- 8 Get Approval & Start Project

FACT FINDING TECHNIQUES

- Interview
- Questionnaire
- Observation
- Sampling of Documents
- Group Discussion
- Site Visit
- Presentation

FEASIBILITY STUDY

- | Technical | Operational | Economic | Legal |
|--------------------------|-----------------------------------|-----------------------|---------------------------|
| Is technology available? | Will system work in organization? | Is it cost effective? | Is it legally acceptable? |

COST-BENEFIT ANALYSIS

- | COSTS | BENEFITS |
|--|---|
| <ul style="list-style-type: none">• Hardware Cost• Software Cost• Development Cost• Training Cost• Operating Cost• Maintenance Cost | <ul style="list-style-type: none">• Increase in Profit• Better Service• Time Saving• Accuracy• Better Control• Competitive Advantage |

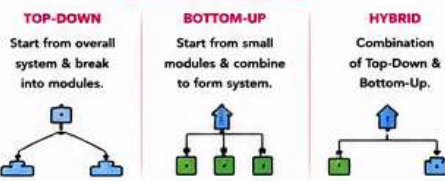
REQUIREMENT GATHERING

- Functional Requirements – What system should do.
- Non-Functional Requirements – How well it should do.

SOURCES: Users, Managers, Existing Documents, Observation, Surveys, Questionnaires

6 MODULAR & STRUCTURED DESIGN + SYSTEM DESIGN & MODELLING

DESIGN APPROACHES



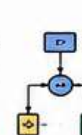
STRUCTURED DESIGN CONCEPTS

- Coupling (Interdependence)
 - Low Coupling (Good)
 - High Coupling (Bad)
- Cohesion (Relatedness)
 - High Cohesion (Good)
 - Low Cohesion (Bad)



DESIGN MODELLING TOOLS

- DFD (Data Flow Diagram)
- Flowcharts
- Structure Charts
- Decision Tables
- Decision Trees
- Pseudo Code



DATA MODELLING

- ER Diagram
- Data Dictionary



KEY POINTS

- Modularity improves maintainability.
- High Cohesion & Low Coupling make good design.
- Logical Design ≠ Physical Design.

7 MORE DESIGN ISSUES, CASE TOOLS, FORMS & REPORTS DESIGN + AUDIT & SECURITY OF COMPUTER SYSTEMS

MORE DESIGN ISSUES

- User Involvement
- Change Management
- Data Security & Integrity
- System Reliability
- Performance & Scalability
- Flexibility & Adaptability
- Usability & Simplicity
- Documentation & Training
- Cost & Time Constraints
- Standards & Compliance



CASE TOOLS (COMPUTER AIDED SOFTWARE ENGINEERING)

Tools to support analysts & designers in SDLC.

BENEFITS

- Increase Productivity
- Ensure Consistency
- Improve Quality
- Reduce Development Time
- Better Documentation
- Easy Maintenance



FORMS DESIGN GUIDELINES

- Clear Title & Purpose
- Simple & Easy to Fill
- Proper Layout & Structure
- Use Proper Labels
- Avoid Unnecessary Fields
- Provide Instructions
- Ensure Accuracy & Readability



REPORTS DESIGN GUIDELINES

- Clear Title & Summary
- Relevant & Accurate Data
- Proper Headings & Subheadings
- Easy to Read & Understand
- Use Tables, Charts & Graphs
- Consistent Format
- Focus on User Needs



AUDIT OF COMPUTER SYSTEMS

OBJECTIVES

- ✓ Ensure Accuracy
- ✓ Ensure Security
- ✓ Ensure Efficiency
- ✓ Ensure Compliance
- ✓ Prevent & Detect Errors
- ✓ Evaluate Effectiveness



SECURITY OF COMPUTER SYSTEMS

CIA TRIAD GOALS

- Confidentiality – Info for authorized users only
- Integrity – Data not altered
- Availability – Info & resources when needed

SECURITY MEASURES

- Access Control
- Encryption
- Firewalls
- Antivirus
- Regular Backups
- Audit Trails



TOP 20 MOST EXPECTED TOPICS

1. SDLC & its Phases
2. Types of Systems
3. System Analyst & Role
4. SRS (System Requirement Spec.)
5. Feasibility Study
6. Cost Benefit Analysis
7. Fact Finding Techniques
8. Documentation Types
9. Top-Down & Bottom-Up Design
10. DFD & its Levels
11. ER Diagram
12. Coupling & Cohesion
13. Data Dictionary
14. Forms vs Reports
15. CASE Tools & Benefits
16. Audit – Types & Objectives
17. Security – CIA Triad
18. Prototype Model
19. JAD (Joint Application Dev.)
20. Logical vs Physical Design

IMPORTANT ABBREVIATIONS

- SDLC – System Development Life Cycle
- SRS – System Requirement Specification
- SDS – System Design Specification
- DFD – Data Flow Diagram
- ERD – Entity Relationship Diagram
- DBMS – Database Management System
- UML – Unified Modeling Language
- CASE – Computer Aided Software Engineering
- JAD – Joint Application Development
- CIA – Confidentiality, Integrity, Availability

ONE-LINER MASTER (ALL UNITS)

- ★ SAD improves system by better analysis & design.
- ★ SDLC provides a structured development path.
- ★ Analyst studies, designs & implements systems.
- ★ Documentation ensures clarity & future reference.
- ★ Planning & feasibility study reduce project risk.
- ★ Modular design makes system easy to maintain.
- ★ CASE tools, forms, reports improve productivity.
- ★ Audit & security protect system from threats.

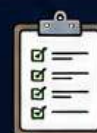


EXAM TIPS (LAST NIGHT)

- ✓ Revise this sheet 2-3 times.
- ✓ Focus on diagrams: SDLC, DFD, ERD, Structure Chart.
- ✓ Write important definitions & points in short notes.
- ✓ Practice PYQs & Top 20 Questions.
- ✓ Stay calm, manage time in exam.
- ✓ Read questions carefully & answer point to point.



PLAN WELL ★ ANALYZE DEEPLY ★ DESIGN SMARTLY ★ DOCUMENT CLEARLY ★ IMPLEMENT SUCCESSFULLY ★ MAINTAIN EFFECTIVELY BEST OF LUCK!



UNIT-1: INTRODUCTION TO SAD

Q. No.	TOP EXPECTED QUESTIONS	ANSWER (Short)	PYQ HINTS
1	Define System.	A system is a set of interrelated components working together to achieve a common goal.	2014, 2016, 2019, 2022
2	What are the characteristics of a System?	Organization, Interaction, Interdependence, Integration, Central Objective.	2015, 2018, 2021, 2023
3	Classify different types of Systems.	Physical, Abstract, Natural, Man-made, Open, Closed, Deterministic, Probabilistic, Simple, Complex.	2014, 2017, 2020, 2024
4	What is Real-Time System? Give example.	System which responds within a fixed time. Example: ATM, Airline Reservation System.	2016, 2019, 2022
5	What is Prototype Model?	A model used to create a quick prototype and refine it based on user feedback.	2015, 2018, 2021, 2023
6	What is JAD? State its advantages.	Joint Application Development involves users, managers & analysts together. Advantages: Better system, User satisfaction, Less rework.	2017, 2020, 2024



UNIT-2: SYSTEMS ANALYST – A PROFESSION

Q. No.	TOP EXPECTED QUESTIONS	ANSWER (Short)	PYQ HINTS
1	Who is a Systems Analyst?	A professional who studies systems, understands user needs and designs effective solutions.	2014, 2016, 2019
2	Explain the role of Systems Analyst.	Study system, analyze requirements, design solutions, evaluate, supervise development, ensure user needs.	2015, 2018, 2021, 2023
3	What are the main skills required by an Analyst?	Technical, Analytical, Communication, Problem solving, Management & Interpersonal skills.	2016, 2019, 2022
4	Why do organizations need Systems Analysts?	To develop better systems, improve efficiency, reduce cost and solve business problems.	2014, 2017, 2020
5	What are the responsibilities of an Analyst?	Gather requirements, prepare SRS, coordinate, test, training, documentation, ensure proper implementation.	2018, 2021, 2023
6	Types of Users.	End User, Managerial User, Technical User.	2015, 2019, 2024



UNIT-3: PROCESS OF SYSTEM DEVELOPMENT

Q. No.	TOP EXPECTED QUESTIONS	ANSWER (Short)	PYQ HINTS
1	Explain SDLC with diagram.	SDLC is a process with phases: Planning → Analysis → Design → Implementation → Maintenance.	Every Year (Repeated)
2	Explain different phases of SDLC.	Planning, Analysis, Design, Implementation, Maintenance.	2014, 2016, 2018, 2022
3	What is the difference between Logical and Physical Design?	Logical Design – What the system should do. Physical Design – How the system will be implemented.	2015, 2019, 2023
4	What is Prototyping Model?	A model where prototype is built, tested and refined as per user feedback.	2016, 2020, 2024
5	What is JAD?	A workshop based approach where users, managers and analysts work together to design system.	2017, 2021, 2023
6	What is Participatory Design?	Users actively participate in the design and development process.	2018, 2022
7	Explain Maintenance phase.	Includes corrective, adaptive, perfective and preventive maintenance.	2014, 2019, 2024



UNIT-4: DOCUMENTATION OF SYSTEMS

Q. No.	TOP EXPECTED QUESTIONS	ANSWER (Short)	PYQ HINTS
1	What is Documentation?	Documentation is recording of all system information in written form.	2014, 2016, 2019
2	List the types of Documentation.	Planning, Requirements, Design, Development, Testing, Operation, Maintenance Documents.	2015, 2018, 2021
3	What is SRS?	System Requirement Specification describes what the system will do and its requirements.	2014, 2017, 2020, 2023
4	What is SDS?	System Design Specification describes how the system will meet the requirements.	2016, 2019, 2022
6	List good documentation practices.	Clear, Accurate, Complete, Consistent, Up-to-date, Easy to understand.	2018, 2022, 2023



UNIT-5: PLANNING AND DESIGNING SYSTEMS

Q. No.	TOP EXPECTED QUESTIONS	ANSWER (Short)	PYQ HINTS
1	What is System Planning?	Planning decides what, how, when and by whom the system will be developed.	2014, 2016, 2019
2	What are Fact Finding Techniques?	Interview, Questionnaire, Observation, Sampling, Group Discussion, Site visit, Presentation.	2015, 2018, 2021, 2023
3	What is Feasibility Study?	It checks viability in terms of Technical, Operational, Economic and Legal feasibility.	2014, 2017, 2020, 2022
4	What is Cost Benefit Analysis?	It compares total cost of system with total benefits to know whether to proceed or not.	2016, 2019, 2023
5	What is Requirement Gathering?	Process of collecting functional and non-functional requirements from users.	2015, 2020, 2024
6	What is the purpose of CBA?	To justify the investment and select the best alternative.	



UNIT-6: MODULAR & STRUCTURED DESIGN + SYSTEM DESIGN & MODELLING

Q. No.	TOP EXPECTED QUESTIONS	ANSWER (Short)	PYQ HINTS
1	What is Top-Down Design?	Start from overall system and decompose into smaller modules.	2014, 2016, 2019
2	What is Bottom-Up Design?	Start from modules and integrate to form the system.	2015, 2018, 2021
3	What is Coupling?	Interdependence between modules.	2014, 2017, 2020
4	What is Cohesion?	Relatedness within a module.	2016, 2019, 2022
5	Draw DFD and explain.	DFD shows flow of data in the system using symbols: Process, Data Flow, Data Store, Source/Sink.	Every Year (Important)
6	What is ER Diagram?	A model that shows entities, attributes and relationships.	2015, 2020, 2023
7	What is Data Dictionary?	Dictionary that stores data details like name, type, size, description etc.	2016, 2019, 2024
8	What is Decision Table?	A table that shows conditions and corresponding actions.	2017, 2021, 2023
9	What is Structure Chart?	A hierarchical chart that shows modules and control relationships.	2014, 2018, 2022

UNIT-7: MORE DESIGN ISSUES, CASE TOOLS, FORMS & REPORTS DESIGN + AUDIT & SECURITY OF COMPUTER SYSTEMS

Q. No.	TOP EXPECTED QUESTIONS	ANSWER (Short)	PYQ HINTS
1	What are CASE Tools?	Tools that support analysts and developers in SDLC. Examples: Computer Aided Software Engineering.	2014, 2016, 2019
2	List benefits of CASE Tools.	Increase productivity, improve quality, reduce development time, better documentation, ensure consistency.	2015, 2018, 2021
3	What are Forms?	Forms are used to collect data from users.	2014, 2017, 2020
4	What are Reports?	Reports present processed information in meaningful form.	2016, 2019, 2022
5	Difference between Form and Report.	Form – Input document to collect data. Report – Output document to present data.	2015, 2020, 2023
6	What are guidelines for good form design?	Simple, Clear, Proper layout, Proper labels, Avoid unnecessary fields, Easy to fill.	2017, 2021, 2024
7	What are guidelines for good report design?	Consistent format, Graphs & tables.	
8	What is Audit?	Audit is a systematic examination of system to ensure accuracy and reliability.	2014, 2018, 2022
9	Objectives of Audit.	Ensure accuracy, ensure security, ensure efficiency, ensure compliance, prevent & detect errors.	2015, 2021, 2024
10	Security of Computer Systems – List measures.	Access control, Passwords, Encryption, Firewalls, Antivirus, Backup, Regular updates, Audit trails.	2016, 2019, 2023

MOST REPEATED TOPICS (2014–2025)

- ★ SDLC & its Phases
- ★ System Analyst – Role & Skills
- ★ Feasibility Study
- ★ Fact Finding Techniques
- ★ SRS & SDS
- ★ DFD & ER Diagram
- ★ Coupling & Cohesion
- ★ Top Down vs Bottom Up
- ★ Cost Benefit Analysis
- ★ CASE Tools
- ★ Forms & Reports
- ★ Audit & Security



QUICK TIPS (EXAM TIME)

- Read question carefully
- Write answers in points
- Draw diagrams wherever needed
- Use examples in answers
- Manage time effectively

IMPORTANT ABBREVIATIONS

SDLC : System Development Life Cycle	CASE : Computer Aided Software Engineering
SRS : System Requirement Specification	JAD : Joint Application Development
SDS : System Design Specification	CBA : Cost Benefit Analysis
DFD : Data Flow Diagram	ROI : Return on Investment
ERD : Entity Relationship Diagram	KPI : Key Performance Indicator
UML : Unified Modeling Language	



PYQ HINTS GUIDE

- 'Every Year' = Very Important (Repeated Frequently)
- 4 or more years = Very High Priority
- 2–3 years = Important
- 1 year = Expected
- Focus on highlighted questions first!



PART-A: SHORT ANSWERS (SYLLABUS TOPICS)

1. PLANNING AND DESIGNING SYSTEMS

- **Fact finding Techniques:** Interviews, Questionnaires, Observation, Sampling of Documents, Group Discussion, Site Visits, Presentations.
- **Feasibility Study:** Technical, Operational, Economic, Legal.
- **Cost Benefit Analysis:** Compares total costs vs total benefits to check profitability.
- **Requirement Gathering:** Collecting functional & non-functional requirement from users.
- **SDLC Phases:** Identification, Planning, Analysis, Design, Implementation, Maintenance.

2. MODULAR AND STRUCTURED DESIGN

- **Design Principles:** Simplicity, Modularity, Abstraction, Information Hiding, Functional Independence.
- **Top Down Design:** Start from overall system and divide into sub-modules.
- **Bottom Up Design:** Start from small modules and combine to form the system.
- **Coupling:** Interdependence between modules (Low is Good).
- **Cohesion:** Relatedness within a module (High is Good).
- **Structure Charts:** Show hierarchical decomposition of modules.

3. SYSTEM DESIGN AND MODELLING

- **Logical Design:** Focus on what the system should do.
- **Physical Design:** Focus on how the system will be implemented.
- **Data Flow Diagram (DFD):** Graphical representation of data flow.
- **Data Modelling:** Defines data, relationships, and constraints.
- **E-R Diagram:** Entity Relationship Diagram shows entities, attributes & relationships.
- **Decision Tables:** Show conditions vs actions in tabular form.
- **Decision Trees:** Show decision logic in tree structure.
- **Notation Structured English:** Uses simple English-like statements for logic.
- **Data Dictionary:** Stores metadata about data items.

4. MORE DESIGN ISSUES AND CASE TOOLS

- **Forms:** Documents used to collect data.
- **Reports:** Documents used to present processed data.
- **Importance of Reports:** Communication, Planning, Control, Decision Making.
- **Differences between Forms & Reports:** Forms collect data, Reports present data.
- **Design Specifications:** Describe what the system will do and how it will perform.
- **Narrative Overviews:** Describe system flow & logic in words.
- **CASE Tools:** Computer Aided Software Engineering tools used for system development.

5. AUDIT AND SECURITY OF COMPUTER SYSTEMS

- **Audit:** Systematic examination of IS to ensure accuracy, efficiency & security.
- **Objectives of Audit:** Ensure accuracy, reliability, efficiency, security, compliance.
- **Types of Audit:** System Audit, Operational Audit, Compliance Audit.
- **Security Goals (CIA Triad):** Confidentiality, Integrity, Availability.
- **Security Measures:** Access Control, Encryption, Firewalls, Antivirus, Backups, Audit Trails.

PART-B: ANALYZING STATEMENTS (Q.1-10) WITH ANSWERS

1	The Earth revolves around the Sun.	→ True. The Earth revolves around the Sun in an elliptical orbit. It takes about 365.25 days to complete one revolution.
2	Python is the best programming language.	→ Subjective. Python is one of the best programming languages due to its simplicity, readability, versatility and vast libraries.
3	This smartphone is the most reliable on the market.	→ Subjective. "Most reliable" depends on user reviews, brand, performance, durability, features and customer satisfaction.
4	Water freezes at 0°C under standard conditions.	→ True. At standard atmospheric pressure (1 atm), water freezes at 0°C (273.15 K).
5	Every student should learn coding.	→ Opinion/True (in general). Coding helps in problem solving, logical thinking and builds a strong career foundation in the digital world.
6	India became independent in 1947.	→ True. India got independence on 15th August 1947 from British rule.
7	Our college has the best faculty in the city.	→ Subjective. Depends on factors like qualifications, experience, teaching quality, student feedback and results.
8	Artificial intelligence is changing business operations.	→ True. AI automates tasks, improves decision making, reduces costs and increases efficiency in almost every industry.
9	Chocolate ice cream tastes better than vanilla.	→ Subjective. Taste is personal; preference varies from person to person.
10	Exercise improves physical health.	→ True. Regular exercise improves cardiovascular health, strengthens muscles, controls weight and reduces stress.

QUICK REVISION – KEY DEFINITIONS (ONE LINERS)

- **System:** A set of interrelated components working together to achieve a goal.
- **SDLC:** System Development Life Cycle.
- **Feasibility Study:** Study to check practicality of a system.
- **DFD:** Shows flow of data in a system.
- **ER Diagram:** Shows entities, attributes and relationships.
- **Coupling:** Interdependence between modules.
- **Cohesion:** Relatedness within a module.
- **Structure Chart:** Hierarchical chart of modules.
- **Data Dictionary:** Details of data items used in system.
- **Documentation:** Recording of all system information.
- **SRS:** Describes what the system will do.
- **SDS:** Describes how the system will be built.
- **Forms:** Used to collect data.
- **Reports:** Used to present data.
- **Audit:** Examination of system for accuracy & security.
- **CIA Triad:** Confidentiality, Integrity, Availability.
- **CASE Tools:** Tools used to develop systems.
- **JAD:** Joint Application Development.



MOST REPEATED QUESTIONS (AKU PYQ HINTS)

- ✓ SDLC & its phases
- ✓ System Analyst – Role & Skills
- ✓ Feasibility Study
- ✓ Fact Finding Techniques
- ✓ SRS & SDS
- ✓ DFD & ER Diagram
- ✓ Coupling & Cohesion
- ✓ Top Down vs Bottom Up
- ✓ Prototype Model
- ✓ CASE Tools
- ✓ Forms vs Reports
- ✓ Audit & Security
- ✓ Logical vs Physical Design



VERY SHORT ANSWERS (EXAM HINTS)

- **Prototype Model:** Build a quick working model and refine it.
- **JAD:** Users, managers & analysts work together.
- **Data Modelling Tools:** ERD, Data Dictionary.
- **Turnaround Document:** Document used to capture output for input.
- **Good Form Design:** Simple, Clear, Accurate, Easy to fill.
- **Good Report Design:** Relevant, Accurate, Timely, Easy to read.
- **Security Measures:** Access control, Encryption, Backup, Audit.

EXAM TIPS (LAST HIGHT)

- ✓ Read Questions Carefully
- ✓ Write Short & Point to Point Answers
- ✓ Use Diagrams / Charts where needed
- ✓ Focus on Definitions & Key Points
- ✓ Practice PYQs (2014-2025)
- ✓ Manage Time – Don't Leave Any Q.
- ✓ Attempt Easy Questions First

